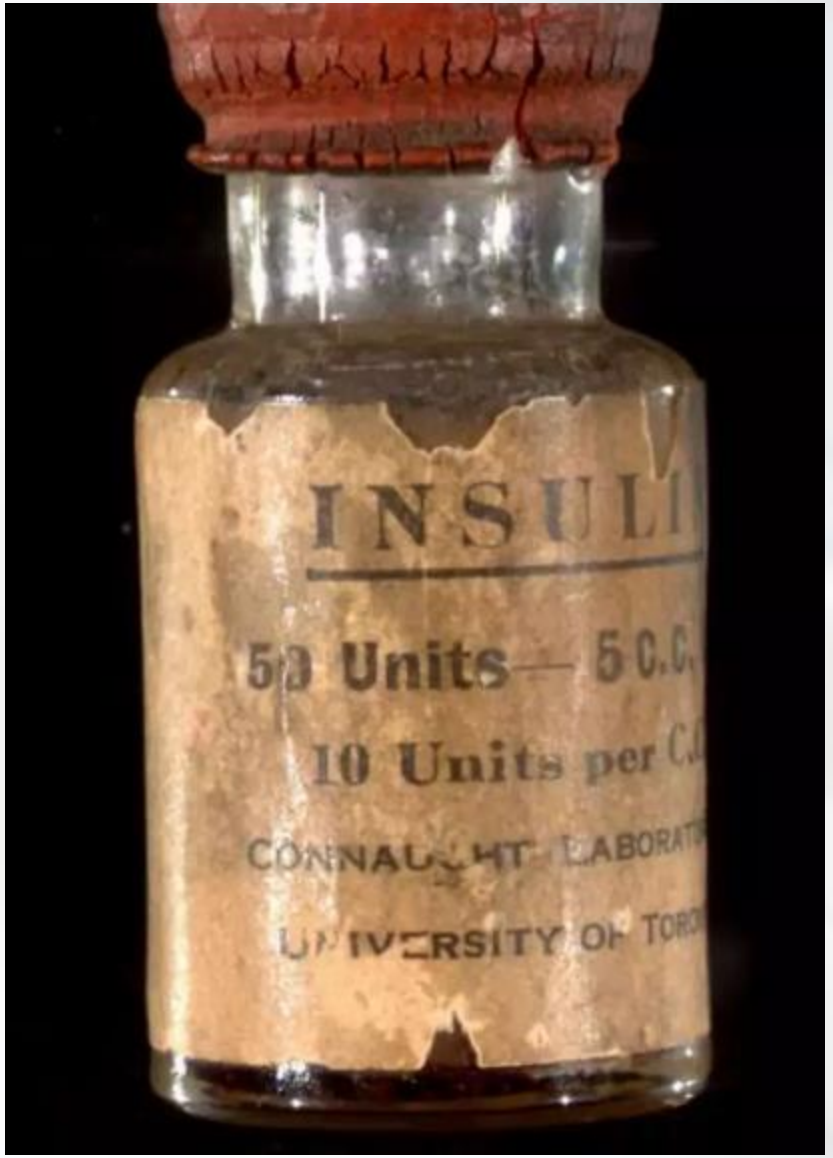
INSULIN THERAPY

DR RAJASHEKHAR MULIMANI
MBBS MD FID
CONSULTANT PHYSICIAN AND DIABETOLOGIST

DR RAJ


FREDERICK GRANT
BANTING
1891 · 1941
JOHN JAMES RICKARD
MACLEOD
1876 · 1935
CHARLES HERBERT
BEST
1899 · 1978
JAMES BERTRAM
COLLIP
1892 · 1965


INSULI
10 Units per C.
UNIVERSITY OF TORO





Patient JL, 15 pounds December 15, 1922



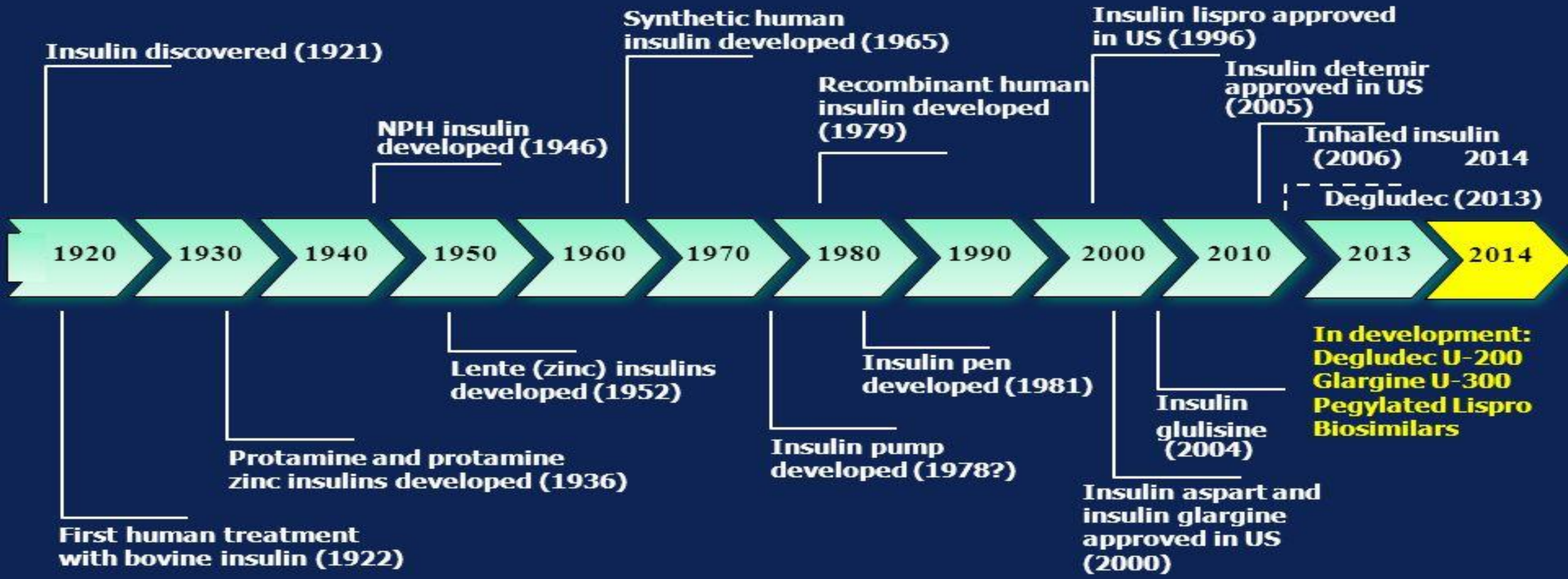
Patient JL, 29 pounds February 15, 1923



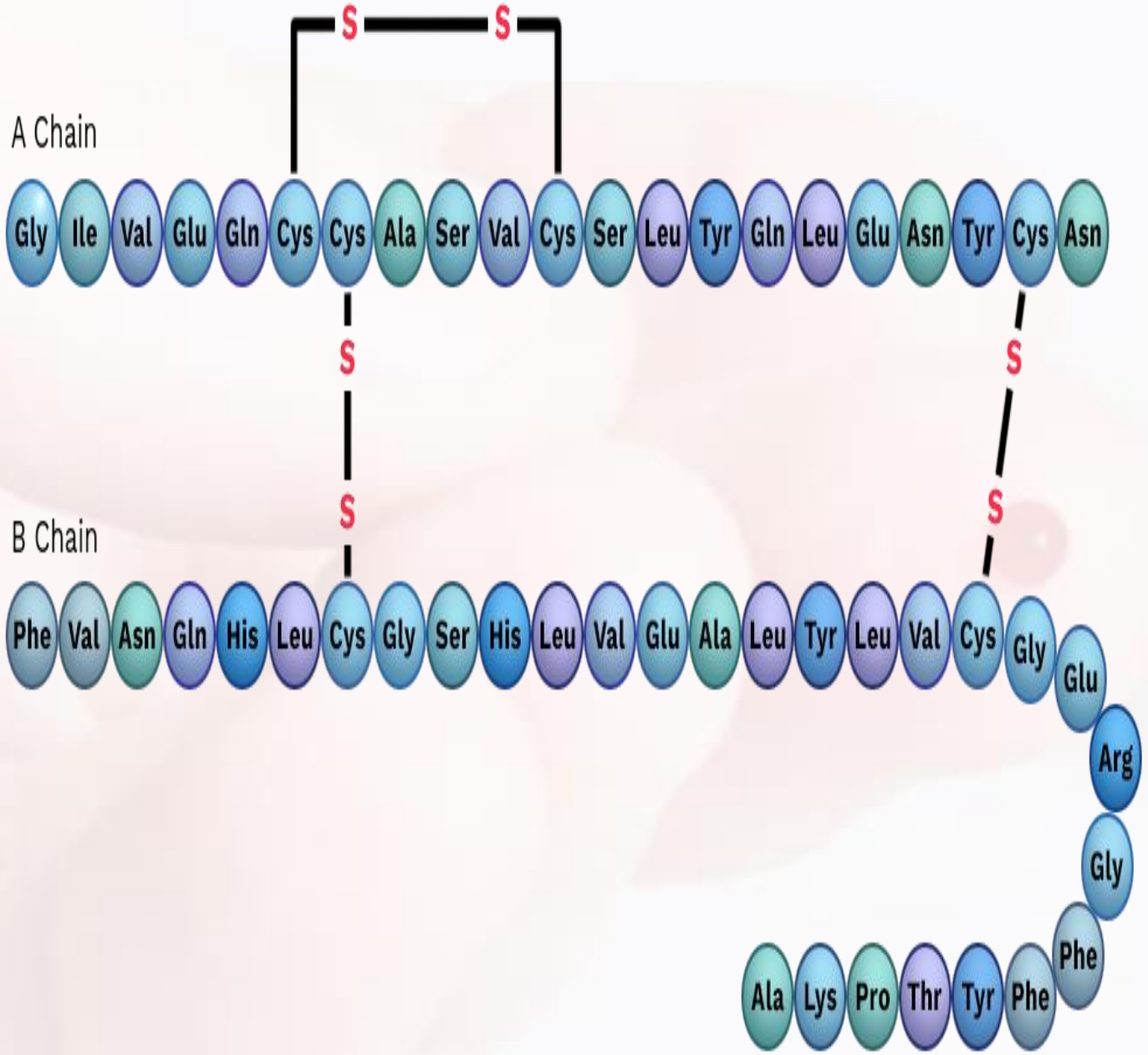
HISTORY MAKERS
(Left) Dr J.P. Bose, the first doctor to use insulin in India on a patient; (right) Kumudini, 12, one of the earliest Indians to receive insulin, in 1929



Milestones in Insulin Development



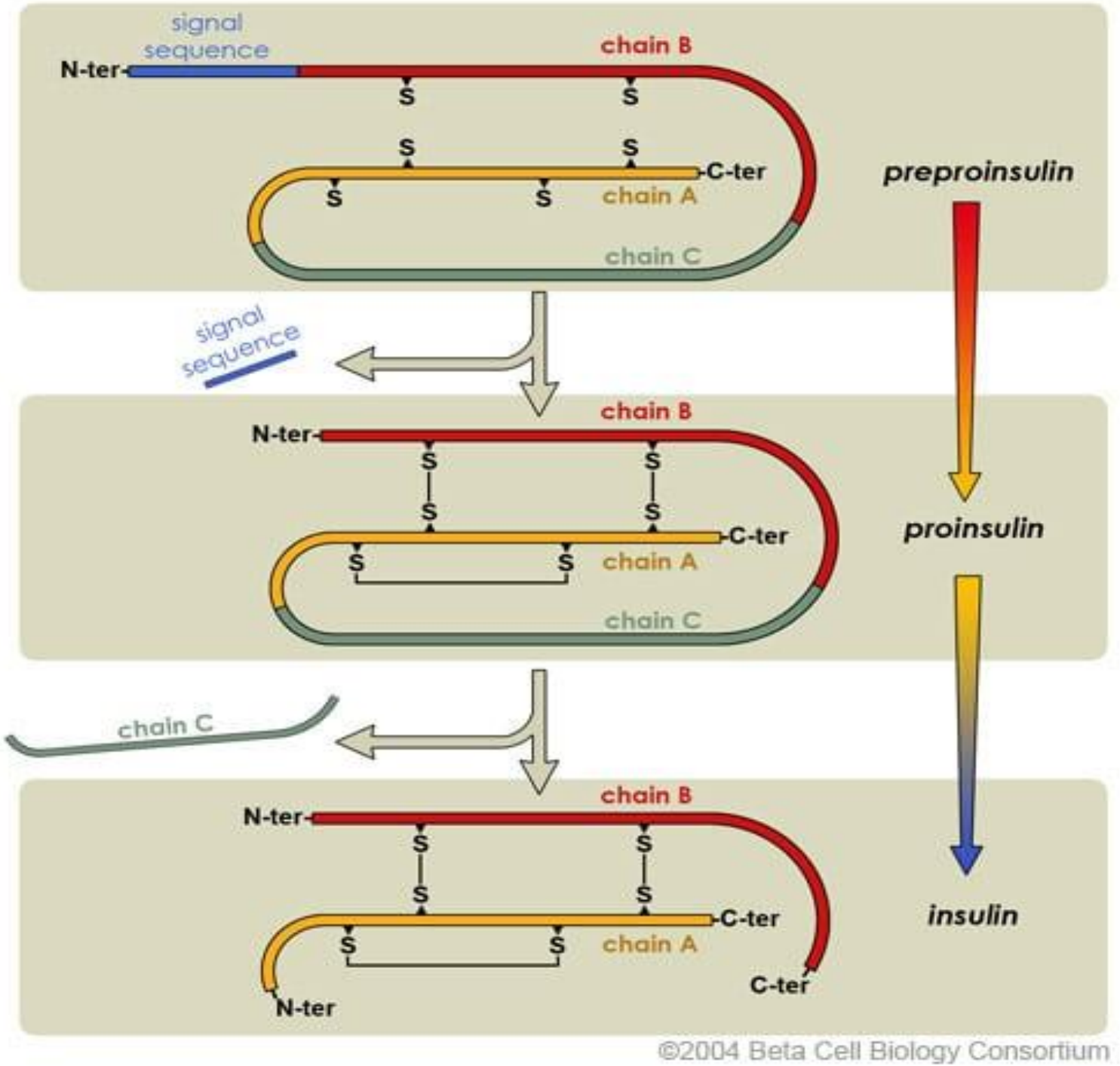
Tattersall RB. In: Pickup JC, Williams G, eds. Textbook of Diabetes. 3rd ed. Blackwell Science: Malden, MA; 2003:1.1-1.22; Drugs@ FDA; http://diabetes.webmd.com/news/20071018/pfizer-quits-inhaled-insulin-exubera.

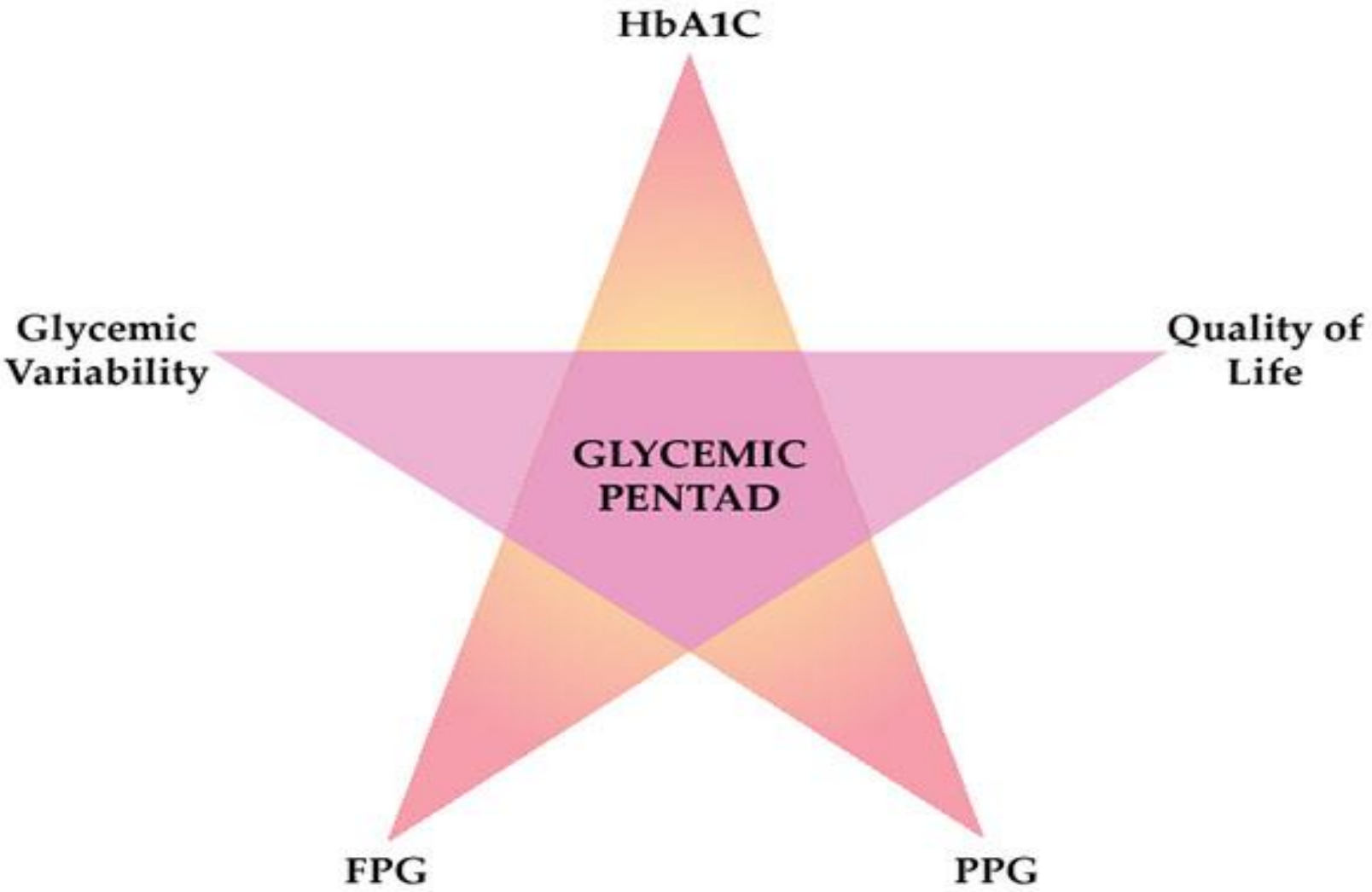


- **HUMAN INSULIN HORMONE IS A PEPTIDE HORMONE.**
- **MADE UP OF 51 AMINO ACIDS, MM OF 5800Da.**
- **DIMER PROTIEN – HAS 2 CHAINS**
  - **A CHAIN – 21 AMINO ACIDS**
  - **B CHAIN – 30 AMINO ACIDS**
- **BOTH CHIANS ARE LINKED TOGETHER BY DISULFIDE BONDS.**
- **STRUCTURE VARIES SLIGHTLY AMONG ANIMALS**
- **CLOSEST TO HUMAN INSULIN IS PORCINE INSULIN**

The initial mRNA products encodes preproinsulin

Proinsulin is a direct precursor of insulin. Gets cleaved in golgi apparatus




HbA1C
Glycemic Variability
Quality of Life
GLYCEMIC PENTAD
FPG
PPG

AVAILABLE INSULIN NOW A DAYS

1.HUMAN INSULIN

- REGULAR INSULIN
- NPH INSULIN
- U-500 REGULAR INSULIN

2.INSULIN ANALOGUES

- RAPID ACTING INSULIN ANALOGUES
- BASAL INSULIN ANALOGUES

3.BIOSIMILIARS

Available formulation of insulins Largely differ in their pharmacokinetics only

By altering its molecular structure

By attaching other moities

At higher concentration at depot site, the insulin molecules form dimers, tertramers or hexamers and then dissolution of insulin molecules must occur before they can gradually cross capillary barrier and gets released into circulation

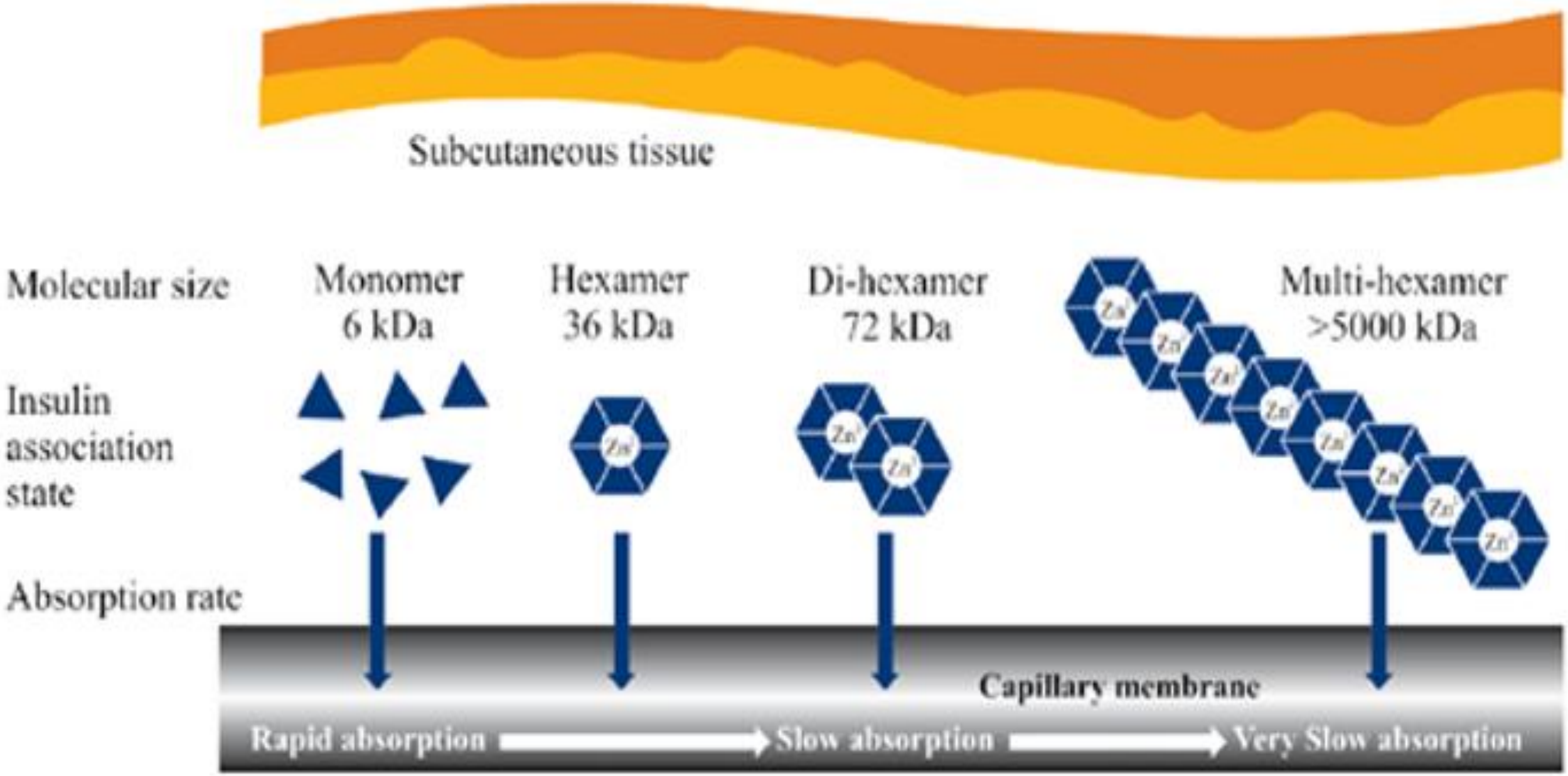


Fig. 1 : Insulin Association State and Absorption Rate.[6]

MAIN PROBLEM IS



REPLACING THE PHYSIOLOGIC PATTERN OF INSULIN SECRETION

- BACKGROUND SECRTION IS LOW, STABLE BETWEEN MEALS, WITH PRANDIAL SURGES.
- ANOTHER IMPORTANT THING HERE IS, IN HEALTHY, ONLY A SMALL PROPORTION OF ENDOGENOUS INSULIN REACHES THE SYSTEMIC CIRCULATION.(LIVER >>>> MUSCLE)
- WHEN INSULIN IS GIVEN EXOGENOUSLY, MAJORITY REACHES THE SYSTEMIC CIRCULATION. (MUSCLE >>>>> LIVER)
- RESULTING AN INABILITY TO SUPRESS THE HEPATIC GLUCOSE SUFFICIENTLY.

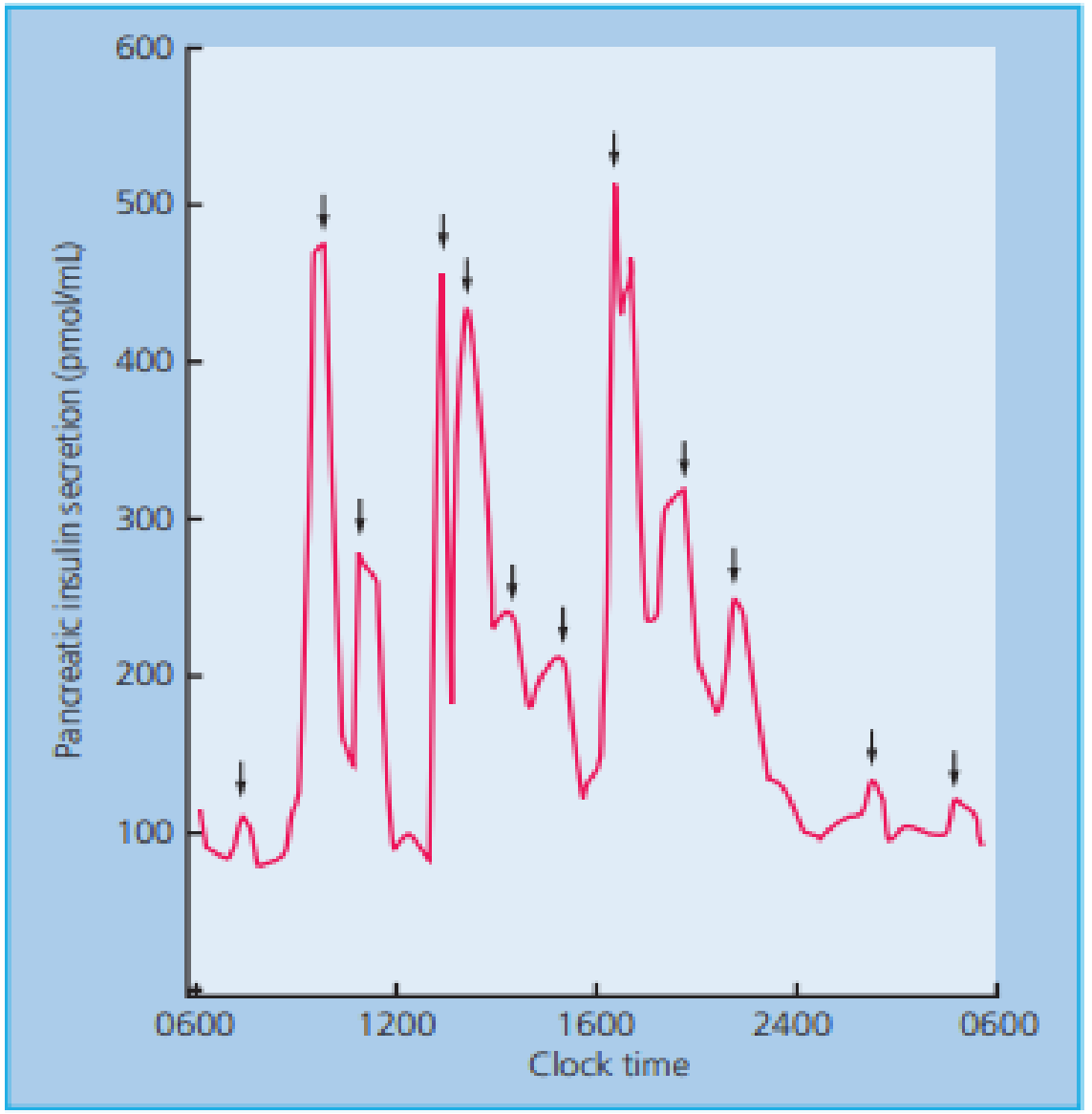


Figure 29.4 Pancreatic insulin secretion in a healthy non-obese individual. Meals were consumed at 0900, 1300, and 1800. Statistically significant pulses of secretion are shown by the arrows. Source: Polansky KS, et al. 1988 [32]. Reproduced with permission from the American Society for Clinical Investigation

CLASSIFICATION OF INSULIN

INSULIN TYPE	ONSET	PEAK	ACTION
MEAL TIME / BOLUS / PRANDIAL INSULIN			
RAPID ACTING ANALOGUES:			
▪ INSULIN LISPRO (HUMALOG)	10-15 MIN	1-2 HR	3-5 HR
▪ INSUIN ASPART (NOVOLOG)	10-15 MIN	1-2 HR	3-5 HR
▪ INSULIN GLULISINE (APIDRA)	10-15 MIN	1-2 HR	3-5 HR
SHORT ACTING:			
▪ REGLUAR INSULIN	0.5 – 1 HR	2-4 HR	4-8 HR
INTERMEDIATE ACTING:			
▪ INSULIN NPH	1-3 HR	4-10 HR	10-18 HR
LONG ACTING ANALOGUES (BASAL INSULIN)			
▪ INSULIN GLARGINE (LANTUS)	2-3 HR	NEGLIGIBLE @ 8 HR	UP TO 24 HOURS
▪ INSULIN GLARGINE U-300 (TOUJEO)	2-3 HR	NONE	UPTO 24 HOUR
▪ INSULIN DETEMIR (LEVEMIR)	2-3 HR	NONE	UPTO 24 HOUR
▪ INSUIN DEGLUDEC (TRESIBA)	2-3 HR	NONE	UPTO 24 HOUR

SHORT ACTING ISNULIN

OR

PRANDIAL INSULIN

OR

MEAL RELATED INSULIN

OR

BOLUS INSULIN

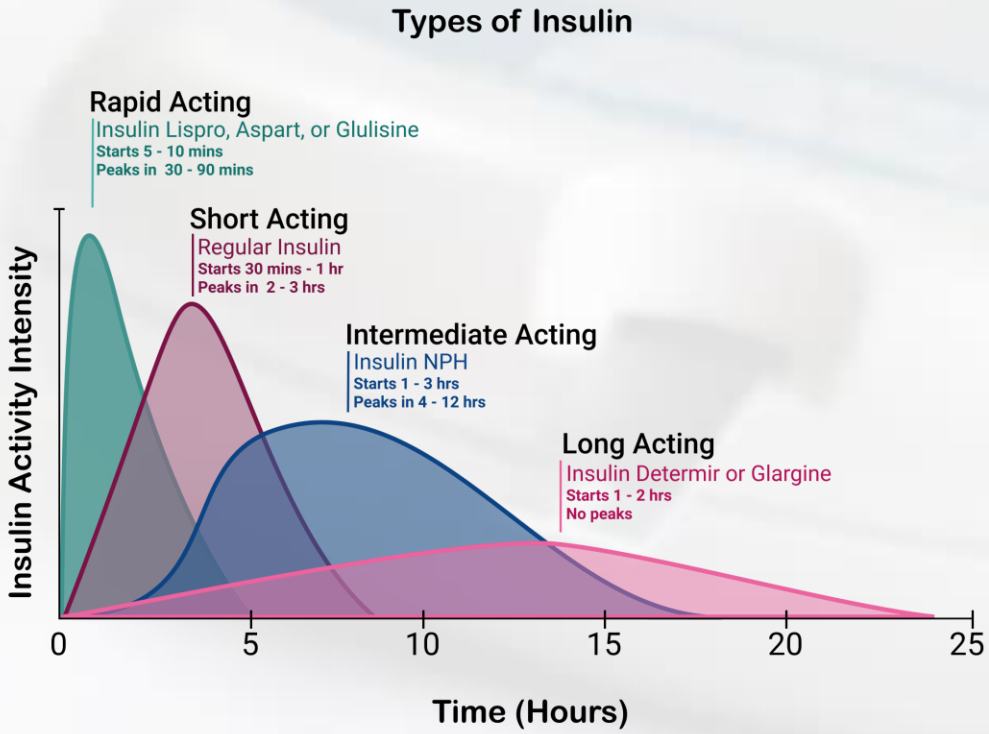
SHORT ACTING ISNULIN
OR
PRANDIAL INSULIN
OR
MEAL RELATED INSULIN
OR
BOLUS INSULIN

1. *REGULAR INSULIN*
2. *RAPID ACTING ANALOGUES (LISPRO/ASPART/GLULISINE)*

REGULAR INSULIN



- IDENTICAL TO ENDOGENOUS INSULIN POLPEPTIDE.
- IN SOLUTION, SELF ASSOCIATES TO FORM HEXAMERS.
- HENCE ACTION STARTS AFTER 30 MINUTES, SO SHOULD BE GIVEN 30 MINS BEFORE MEALS.
- IT DOES NOT MATCH WITH THE MEAL STIMULATED ENDOGENOUS INSULIN SECRETION. (DELAYED PEAK AND SLIGHT LONGER DURATION)
- ***THEREFORE, ISSUES ARE***
  - ✓ ***EARLY POSTPRANDIAL HYPERGLYCEMIA***
  - ✓ ***LATE HYPOGLYCEMIA.***




Humulin® R U-500
500
Warning – Highly Concentrated
U-500

RAPID ACTING ANALOGS



- **UNLIKE, REGULAR INSULIN, THEY ARE ABSORBED RAPIDLY AFTER SC INJECTION, BECAUSE THEY DISSOCIATES INTO MONOMERS ALMOST INSTANTANEOUSY**



- **INJECTED 15 MINS OR LESS BEFORE A MEAL.**

- **HENCE ACTION STARTS AFTER 10-15 MINUTES, AND PEAK AT 2 HOURS.**



- **ADVANTAGES:**
  - ✓ **LOWER RATES OF HYPOGLYCEMIA**
  - ✓ **BETTER POSTMEAL CONTROL**
  - ✓ **SOMETIMES CAN BE GIVEN AFTER MEAL ALSO**
  - ✓ **WHEN GIVEN IV, YOU ARE JUST INCREASING THE COST**

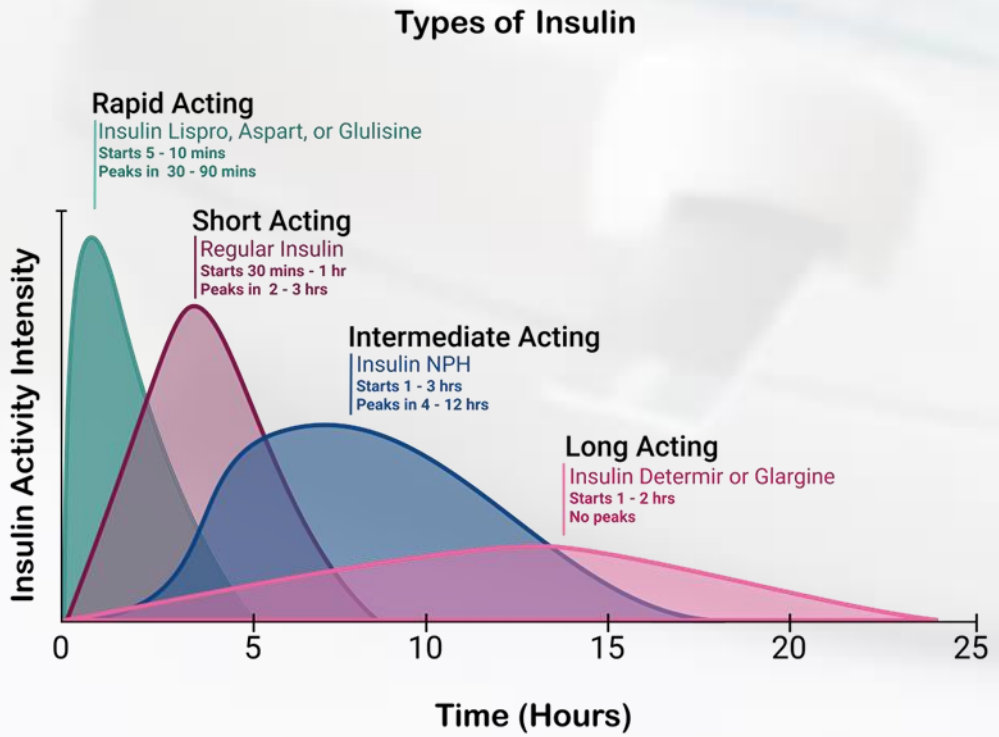
BASAL INSULINS
OR
LONG ACTING INSULINS

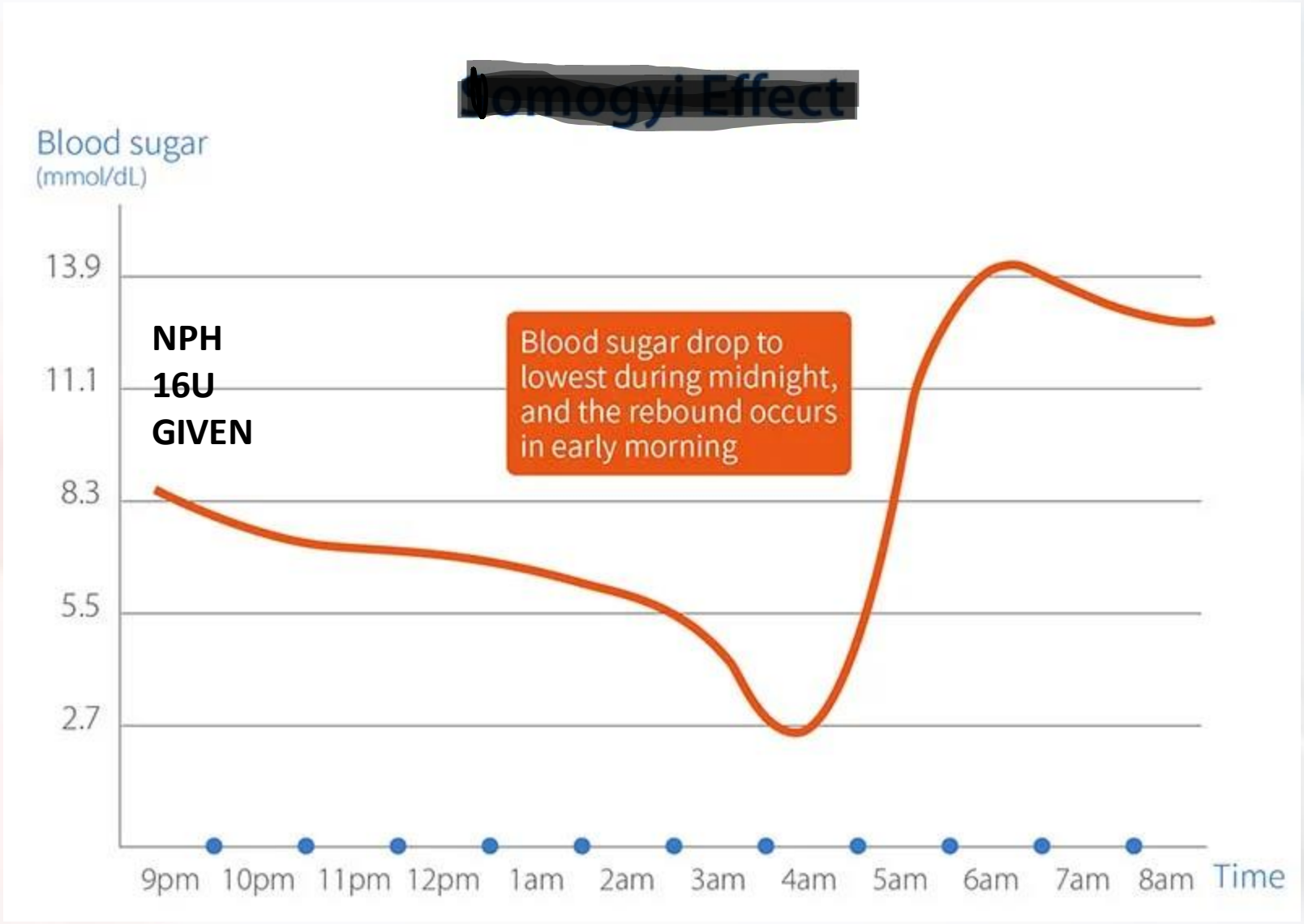
1. *INSULIN NPH*
2. *LONG ACTING ANALOGUES (GLARGINE/DETEMIR/DEGLUDEC)*

INSULIN NPH



- IT'S A SUSPENSION OF INUSLIN COMPLEXES WITH PROTAMINE AND ZINC IN A PHOSPHATE BUFFER.
- THIS PROPERT OF NPH MAKES IT VERY GRADUALLY ABSORBED FROM SC TISSUE.
- CLOUDY INSULIN.
- POPULAR IN INDIA DUE TO LESSER PRICE.
- ONCE OR TWICE DAILY.




Somogyi Effect
Blood sugar
(mmol/dL)
13.9
11.1
8.3
5.5
2.7
NPH
16U
GIVEN
Blood sugar drop to lowest during midnight, and the rebound occurs in early morning
9pm
10pm
11pm
12pm
1am
2am
3am
4am
5am
6am
7am
8am
Time

INSULIN GLARGINE



- **ONLY ACIDIC INSULIN, AND HENCE CANNOT BE MIXED WITH ANY OTHER INSULIN.**
- **FORMS AGGERGATES IN SC TISSUE WITH SLOW (NEAR 24 HOUR DISSOCIATION)**
- **HENCE OD DOSAGE IS POSSIBLE.**
- **SOME HAVE CLAIMED IT AGAINST THE TRUE PEAKLESS EFFECT.**
- **HENCE A DIFFERENT GLARGINE CAME INTO MARKET…. U-300 (TOUJEO)**



New insulin glargine 300 units/mL versus glargine 100 units/mL in people with type 2 diabetes using basal and mealtime insulin: glucose control and hypoglycemia in a 6-month randomized controlled trial (EDITION 1)

Matthew C Riddle [1], Geremia B Bolli [2], Monika Ziemen [3], Isabel Muehlen-Bartmer [3], Florence Bizet [4], Philip D Home [5]; EDITION 1 Study Investigators

INSULIN DETEMIR

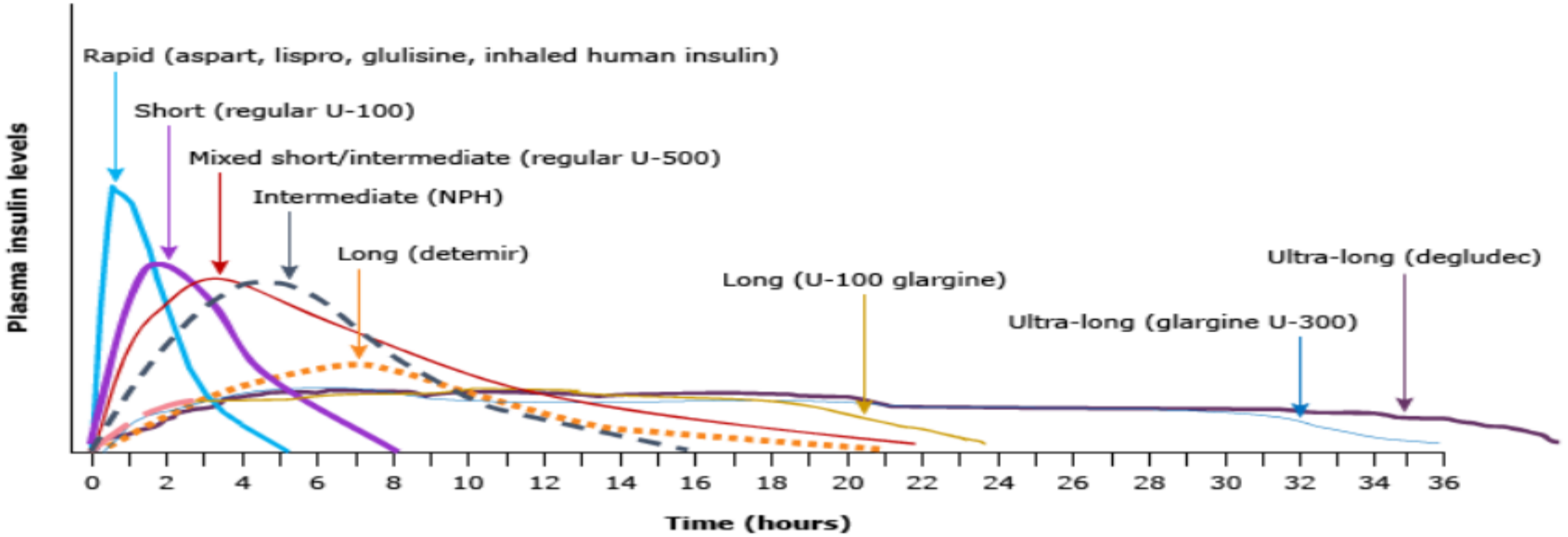
- **ALONG WITH CHANGE IN AMINO ACID, A FATTY ACID GROUP IS ADDED TO BETA CHAIN.**
- **WHICH MAKES IT BINDING POSSIBLE WITH ALBUMIN IN SC TISSUE.**
- **THIS ALBUMIN BINDING DELAYS SC ABSORPTION.**
- **DETEMIR OFTEN REQUIRES TWICE DAILY DOSING.**

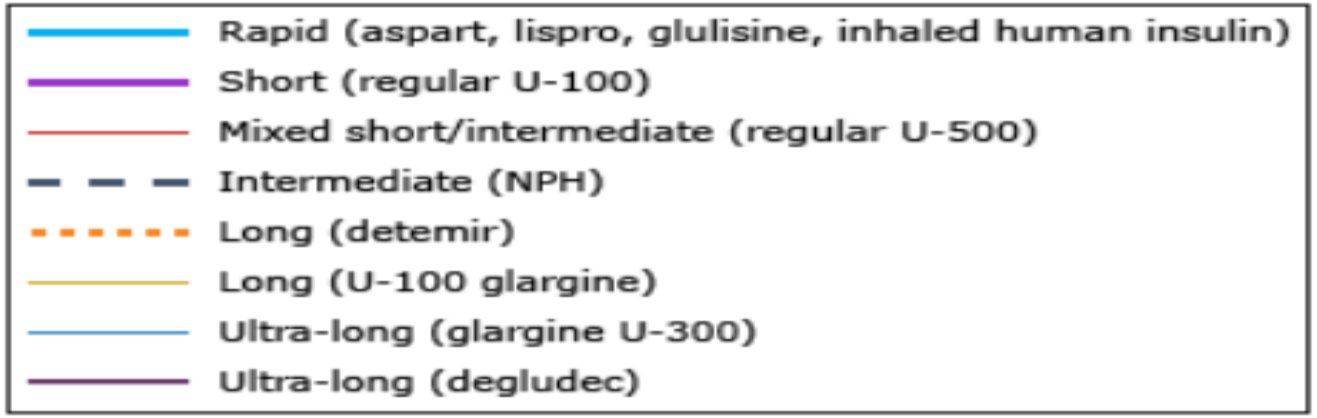


INSULIN DEGLUDEC

- AN ULTRA-LONG ACTING INSULIN.
- FORMS MULTI-HEXAMER AGGREGATES IN SC TISSUE.
- HALF LIFE IS 25.4 HOURS.
- PEAKLESS BASAL PROFILE, WITH DURATION OF ACTION EXTENDING UPTO 42 HOURS.
- IT CAN BE INJECTED ONCE IN 24 HOURS WITH FLEXIBILITY OF VARIABLE TIMINGS




Rapid (aspart, lispro, glulisine, inhaled human insulin)
Short (regular U-100)
Mixed short/intermediate (regular U-500)
Intermediate (NPH)
Long (detemir)
Long (U-100 glargine)
Ultra-long (glargine U-300)
Ultra-long (degludec)
Plasma insulin levels
0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36
Time (hours)


Rapid (aspart, lispro, glulisine, inhaled human insulin)
Short (regular U-100)
Mixed short/intermediate (regular U-500)
Intermediate (NPH)
Long (detemir)
Long (U-100 glargine)
Ultra-long (glargine U-300)
Ultra-long (degludec)

PRE MIXED INSULIN

CONVENTIONAL PREMIXED INSULINS	PREMIXED INSULIN ANALOGUES
30/70 (30% REGULAR + 70% NPH)	HUMALOG MIX 25/75 (25%LISPRO + 75%LISPRO PROTAMINE
50/50 (50% REGULAR + 50% NPH)	HUMALOG MIX 50/50 (50%LISPRO + 50% LISPRO PROTAMINE 
25/75 (25% REGULAR + 75% NPH)	NOVOMIX 30/70 (30%ASPART + 70%ASPART PROTAMINE) 
	RYZODEG 30/70 (30%ASPART + 70% DEGLUDEC) 

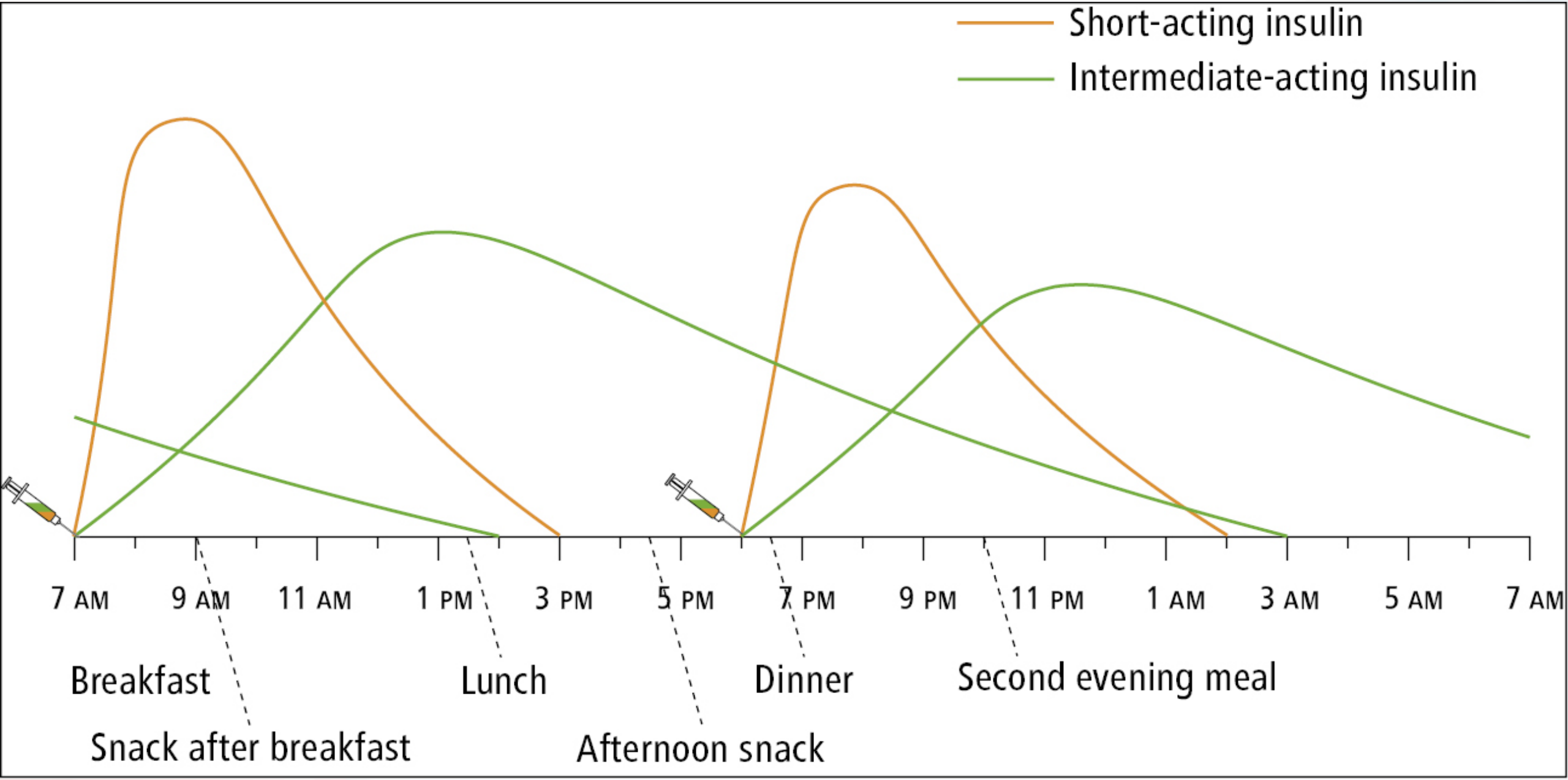
PRE-MIXED INSULIN

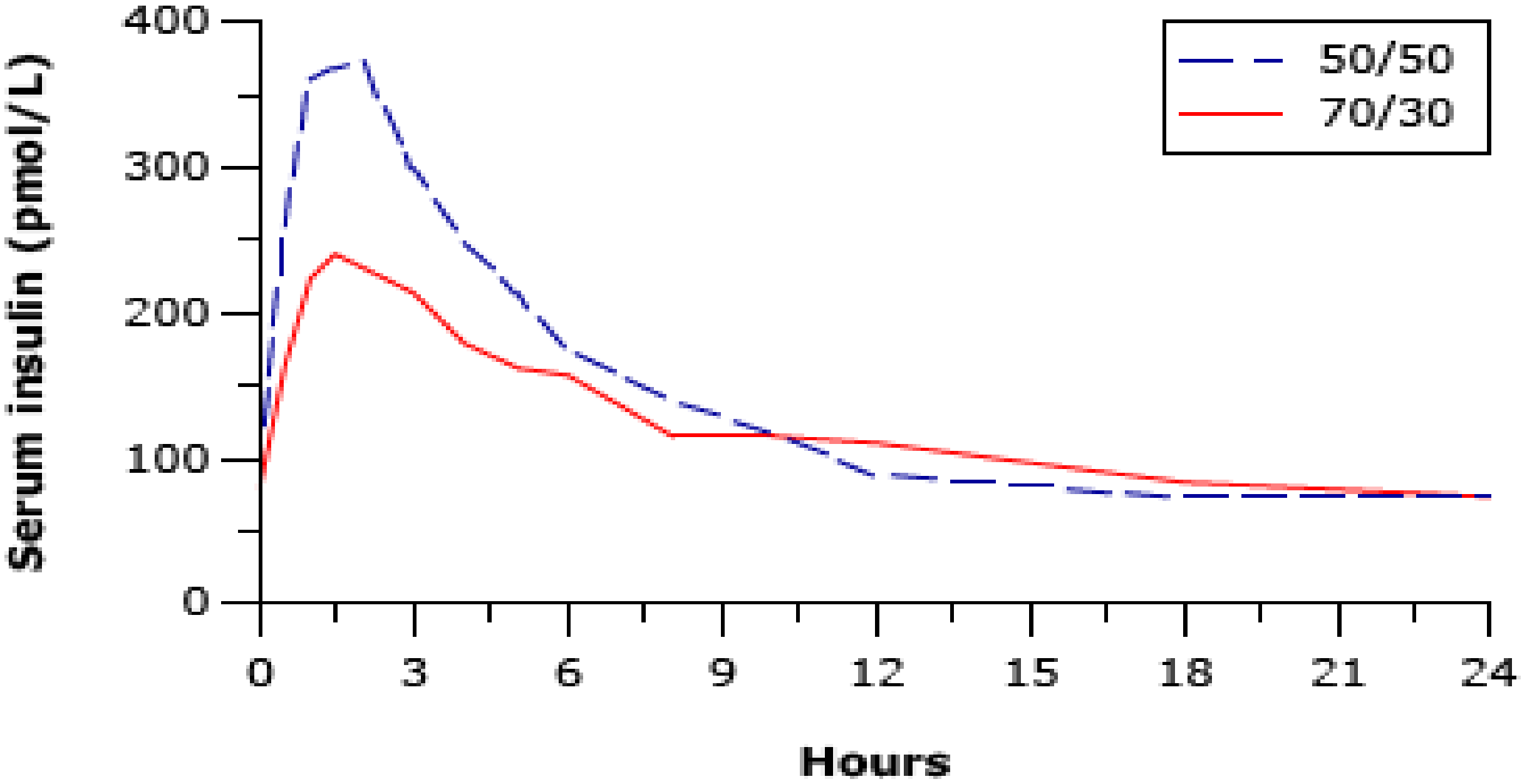
ADVANTAGES

- **LESSER INJECTIONS**

DISADVANTAGES

- **FINE TUNING NOT POSSIBLE**
- **STRICT MEAL PATTERN**
- **NOCTURNAL HYPOGLYCEMIA**
- **MAY NEED “DIET CHANGES FOR INSULIN” rather than “INSULIN CHANGES FOR DIET”**


Short-acting insulin
Intermediate-acting insulin
7 AM
9 AM
11 AM
1 PM
3 PM
5 PM
7 PM
9 PM
11 PM
1 AM
3 AM
5 AM
7 AM
Breakfast
Snack after breakfast
Lunch
Afternoon snack
Dinner
Second evening meal


Serum insulin (pmol/L)
400
300
200
100
0
0
3
6
9
12
15
18
21
24
Hours
50/50
70/30

INDICATIONS FOR INSULIN

1. **TYPE 1 DIABETES MELLITUS**

2. **SECONDARY DIABETES (PANCREATITIS, STEROID INDUCED)**

3. **GESTATIONAL DIABETES MELLITUS**

4. **ACUTE DIABETIC COMPLICATIONS**

5. **TYPE 2 DIABETES MELLITUS**
   - ***Primary or secondary OHA failure.***
   - ***During acute stress.***
   - ***Chronic renal failure***
   - ***Use of parenteral nutrition***
   - ***Weight loss of more than 10% body weight***

THERAPEUTIC INSULIN REGIMENS

AUGMENTATION THERAPY

- WHEN GLYCEMIC FAILURE ON ORAL MEDICATIONS.
- RESIDUAL BUT INSUFFICIENT BETA CELL FUNCTION.
- EITHER BASAL OR BOLUS DEPENDING ON FASTING OR POSTPRANDIAL HYPERGLYCEMIA RESPECTIVELY.
- USUALLY AT 0.3 U/KG

REPLACEMENT THERAPY

- WHEN PATIENT NEEDS INTENSIVE CONTROL OR IN THOSE WHO HAVE FAILED AUGMENTATION THERAPY.
- (BASAL-BOLUS REGIMEN) + CORRECTION
- 0.6 TO 1 U/KG

INSULIN REGIMENS FOR T1DM

1. CONVENTIONAL REGIMEN

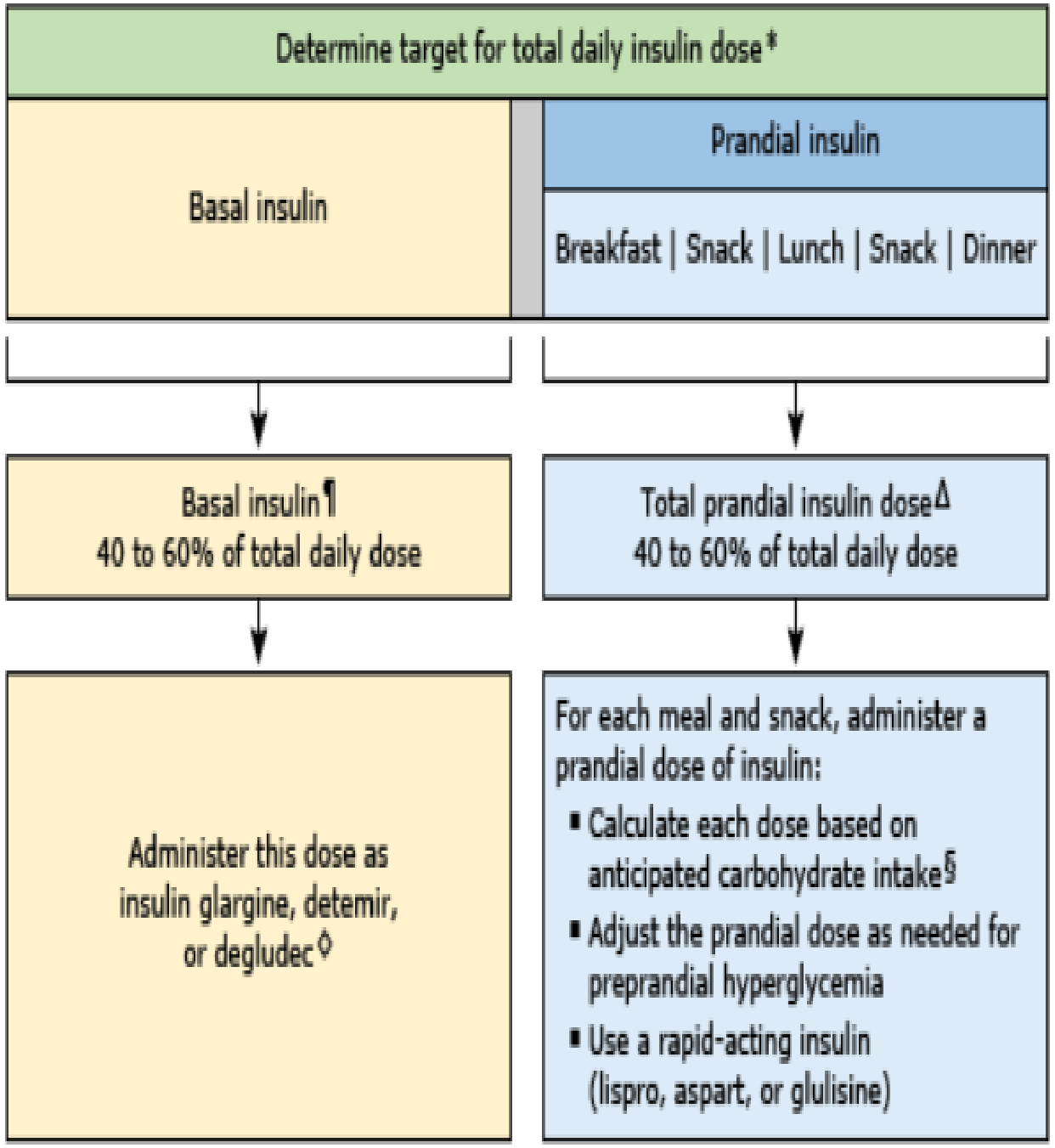
FIXED DOSES OF NPH TWICE DAILY
+
FIXED DOSE SHORT ACTING INSULIN TWICE OR THRICE DAILY

2. INTENSIVE INSULIN REGIMEN

BASAL – BOLUS REGIMEN
CSII – CONTINOUS SUBCUTANEOUS INSULIN INFUSION

3. MODIFIED INSULIN REGIMEN

@ BREAKFAST – INSULIN NPH + RAPID ACTING INSULIN.
@DINNER --------INSUIN GLARGINE/DETEMIR + RAPID ACTING INSULIN



- Total daily insulin dose for children with newly diagnosed type 1 diabetes mellitus generally is between 0.3 and 1 units/kg. Prepubertal children require doses on the lower end of this range (0.3 to 0.7 units/kg/day). Higher doses (eg, 0.7 to 1 units/kg/day) are needed in pubertal children or patients in ketoacidosis or on corticosteroids and, sometimes, for those with obesity. These doses are titrated up or down as needed depending on the patient's response.

- The total prandial insulin dose may vary each day, depending on food intake and any dose corrections for blood glucose above the target range.

- Basal insulin is usually given once daily and usually in the evening. However, some children do better with 2 divided doses, eg, those who tend to have nocturnal hypoglycemia or when detemir insulin is used. Insulin degludec may also be used; it has a longer duration of action but is still given once daily.

INSULIN REGIMENS FOR T2DM

IN EXCESS OF 50% OF PEOPLE WITH T2DM WILL REQUIRE INSULIN INJECTIONS AT SOME POINT IN THEIR LIFE

INSULIN REGIMENS FOR T2DM

1.FIRST LINE REGIMEN.

- *BASAL ONLY REGIMEN*

2.SECOND LINE REGIMEN.

- *PREMIXED REGIMENS, TWICE OR THRICE DAILY*
- *BASAL + MEAL-TIME INSULIN*
- *BASAL-BOLUS REGIMEN*

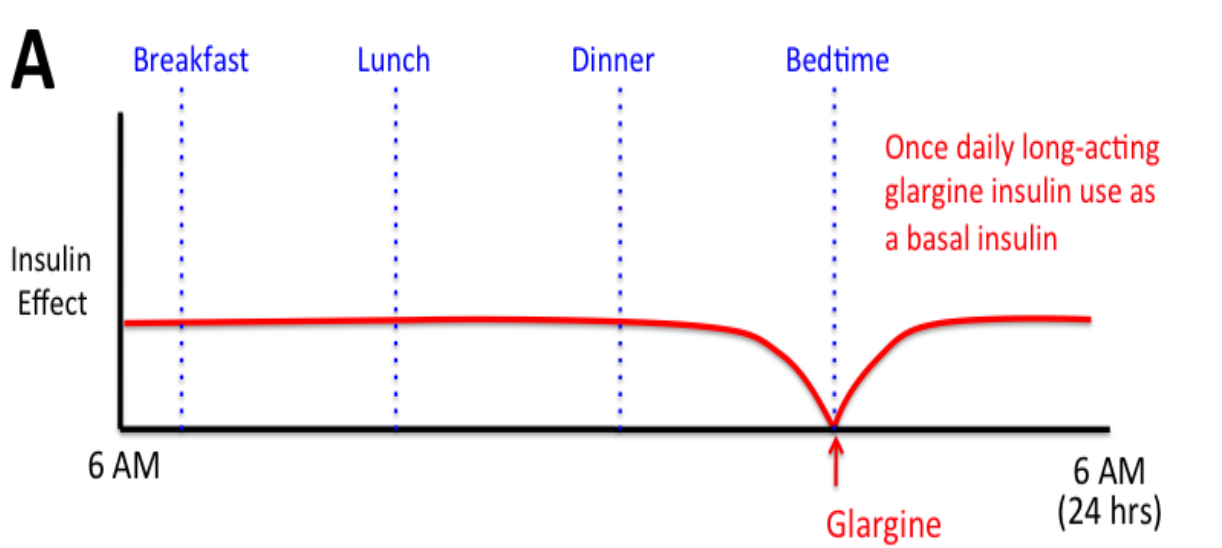
INSULIN REGIMENS FOR T2DM

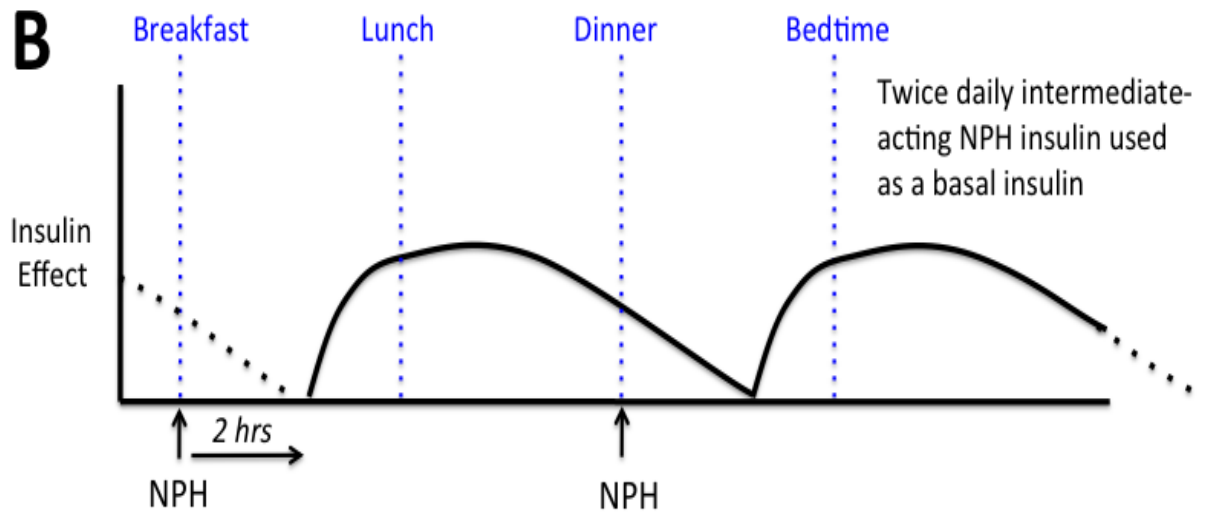
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- *BASAL + MEAL-TIME INSULIN*
- *BASAL-BOLUS REGIMEN*





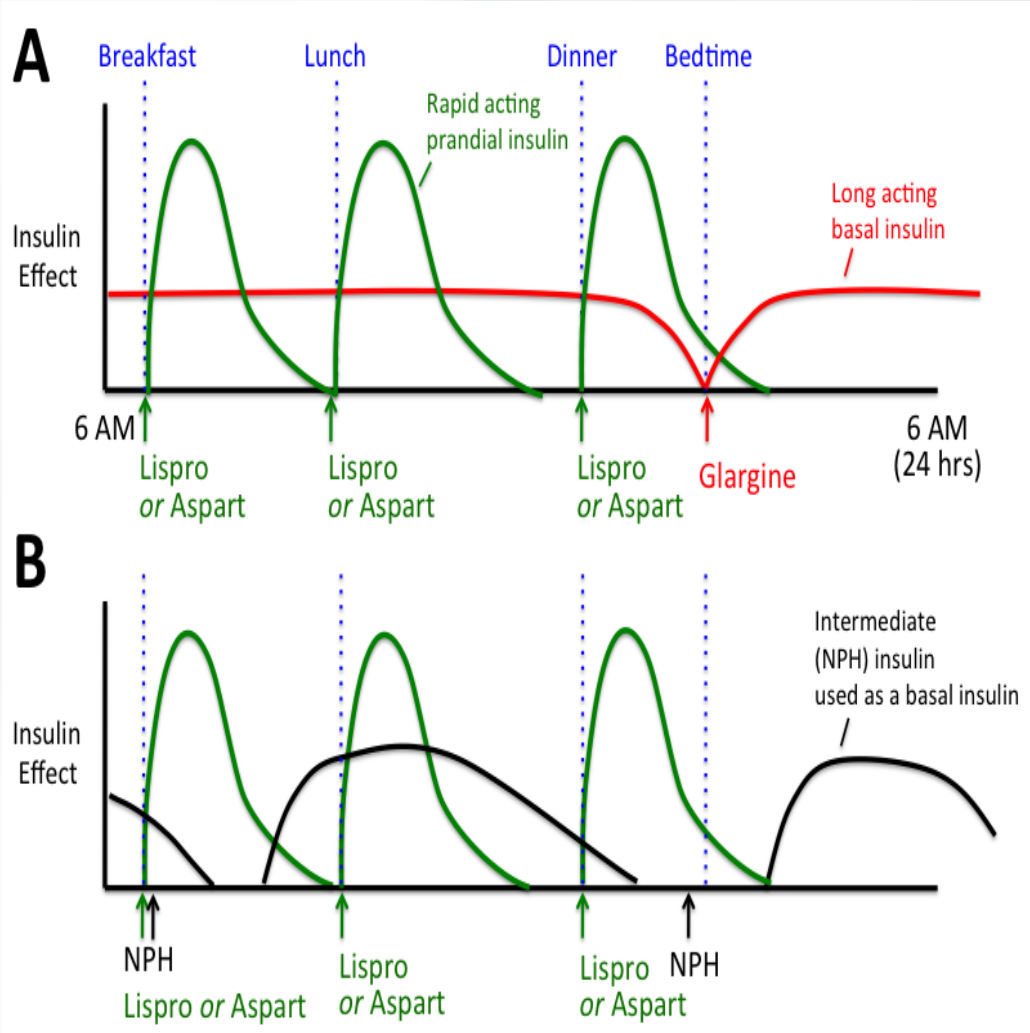
INSULIN REGIMENS FOR T2DM

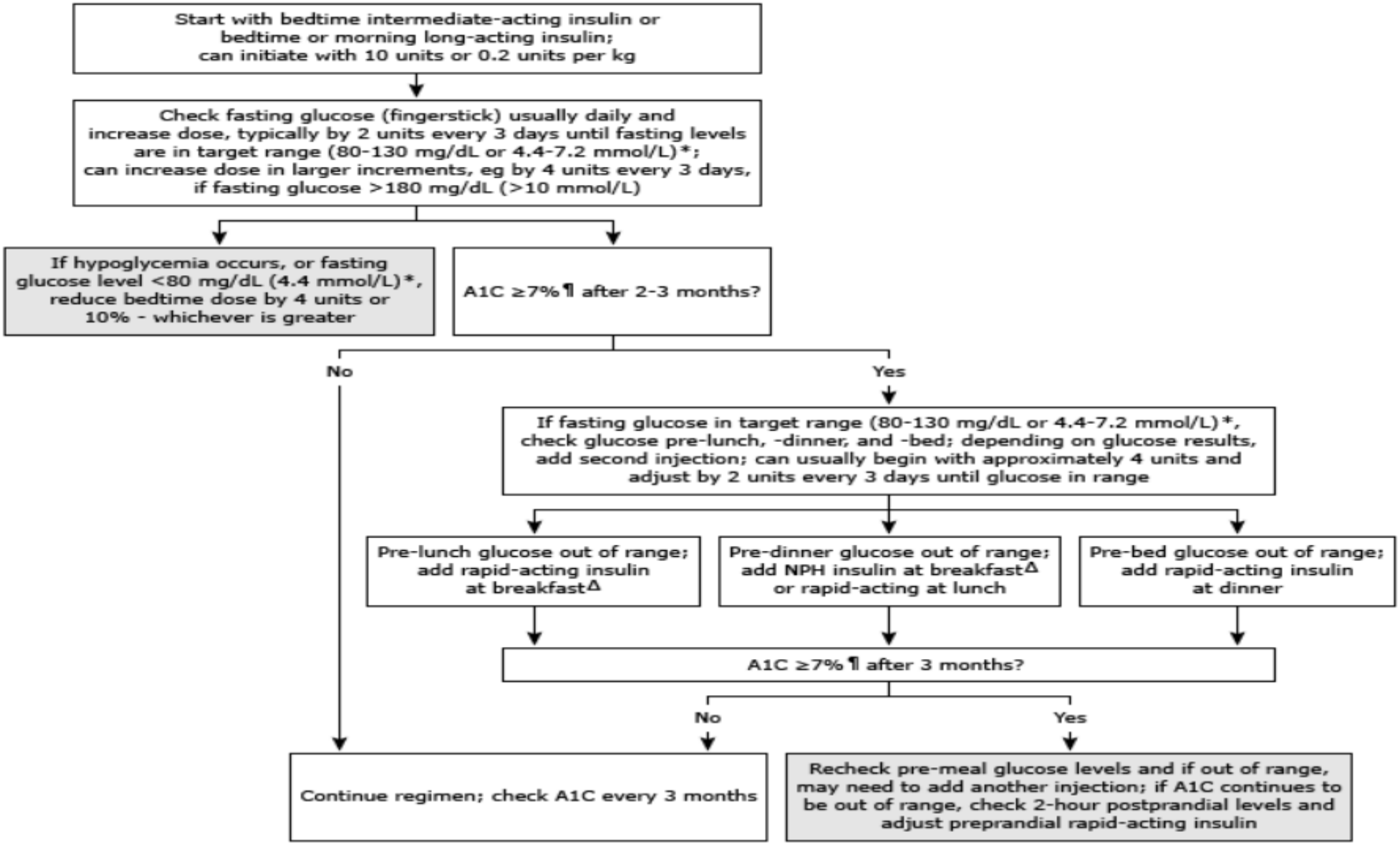
1.FIRST LINE REGIMEN.

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2.SECOND LINE REGIMEN.

- *PREMIXED REGIMENS, TWICE OR THRICE DAILY*
- *BASAL + MEAL-TIME INSULIN*
- *BASAL-BOLUS REGIMEN*




Start with bedtime intermediate-acting insulin or bedtime or morning long-acting insulin; can initiate with 10 units or 0.2 units per kg
Check fasting glucose (fingerstick) usually daily and increase dose, typically by 2 units every 3 days until fasting levels are in target range (80-130 mg/dL or 4.4-7.2 mmol/L)*; can increase dose in larger increments, eg by 4 units every 3 days, if fasting glucose >180 mg/dL (>10 mmol/L)
If hypoglycemia occurs, or fasting glucose level <80 mg/dL (4.4 mmol/L)*, reduce bedtime dose by 4 units or 10% - whichever is greater
A1C ≥7%¶ after 2-3 months?
No
Yes
If fasting glucose in target range (80-130 mg/dL or 4.4-7.2 mmol/L)*, check glucose pre-lunch, -dinner, and -bed; depending on glucose results, add second injection; can usually begin with approximately 4 units and adjust by 2 units every 3 days until glucose in range
Pre-lunch glucose out of range; add rapid-acting insulin at breakfastΔ
Pre-dinner glucose out of range; add NPH insulin at breakfastΔ or rapid-acting at lunch
Pre-bed glucose out of range; add rapid-acting insulin at dinner
A1C ≥7%¶ after 3 months?
No
Yes
Continue regimen; check A1C every 3 months
Recheck pre-meal glucose levels and if out of range, may need to add another injection; if A1C continues to be out of range, check 2-hour postprandial levels and adjust preprandial rapid-acting insulin

TROUBLESHOOTING

- **HYPOGLYCEMIA**
- **INSUIN RESISTANCE**
- **INSULIN INDUCED WIEGHT GAIN**

- **LESSER IN TYPE 2 THAN IN TYPE 1.**
- **BASAL INSULIN LESSER THAN PRANDIAL INSULIN**
- **MAY LEAD TO OBLIGATE EATING**
- **NOCTURNAL HYPOGLYCEMIA**
  - ✓ ***If on NPH then change to long acting analoges***
  - ✓ ***Reduce night time basal insulin by 4U or 10%.***
  - ✓ ***If no precipitant is observed, greater dose reduction like 20-30% can be considered.***
- **DAYTIME HYPOGLYCEMIA**
  - ✓ ***If the patient is taking prandial insulin, the dose should be decreased at the appropriate meal(s)***

TROUBLESHOOTING

- **HYPOGLYCEMIA**
- **INSUIN RESISTANCE**
- **INSULIN INDUCED WIEGHT GAIN**

- In patients with severe insulin resistance (eg, requiring more than 200 total units of insulin daily), concentrated insulins can be used to manage hyperglycemia.
- The smaller volume of injected insulin improves the absorption kinetics.
- ***OPTIONS INCLUDE***
  1. ***U-500 REGULAR INSULIN***
  2. ***U-300 GLARGINE***
  3. ***U-200 LISPRO & U-200 LISPRO aabc***

TROUBLESHOOTING

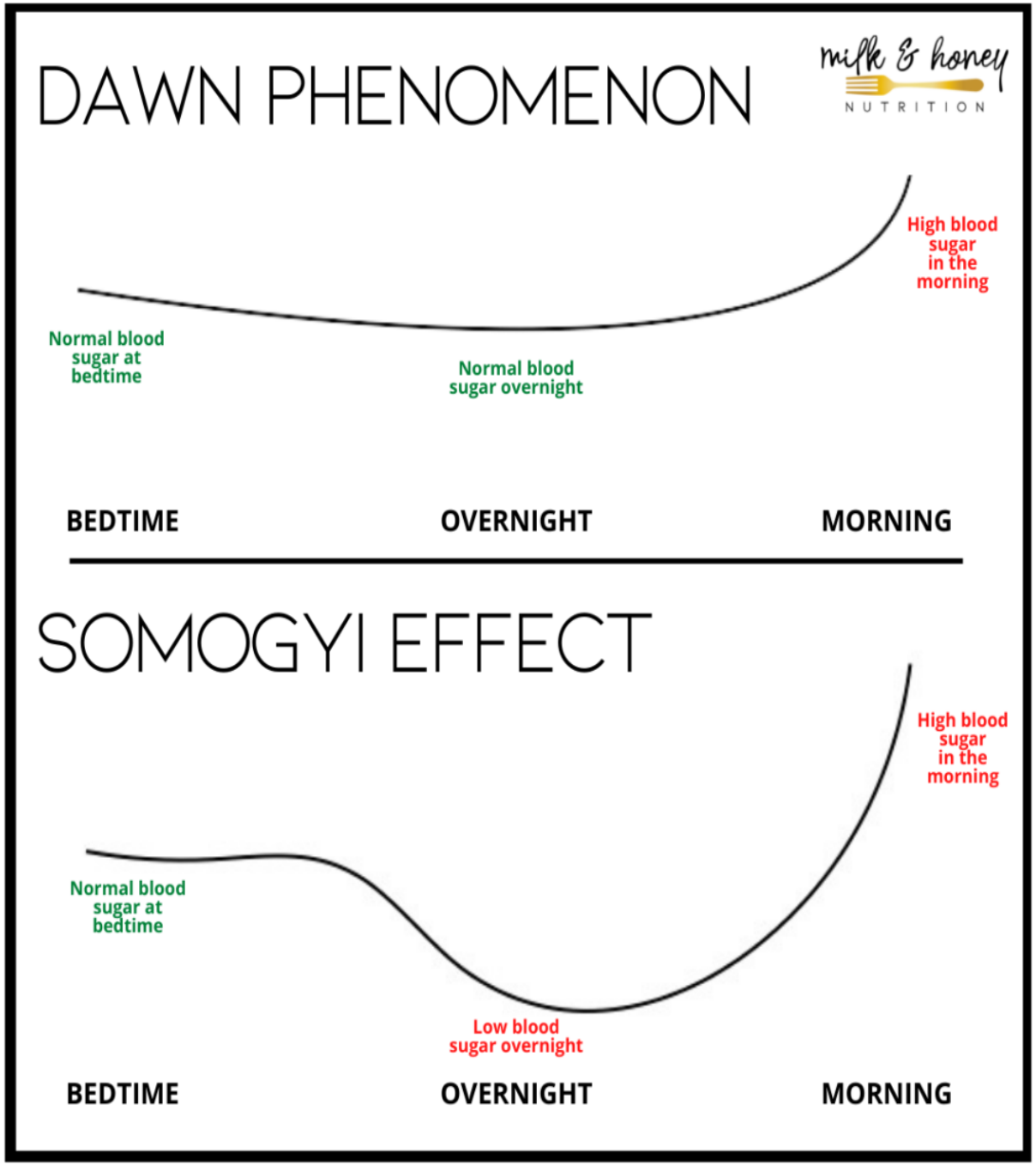
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- **INSUIN RESISTANCE**
- **INSULIN INDUCED WIEGHT GAIN**

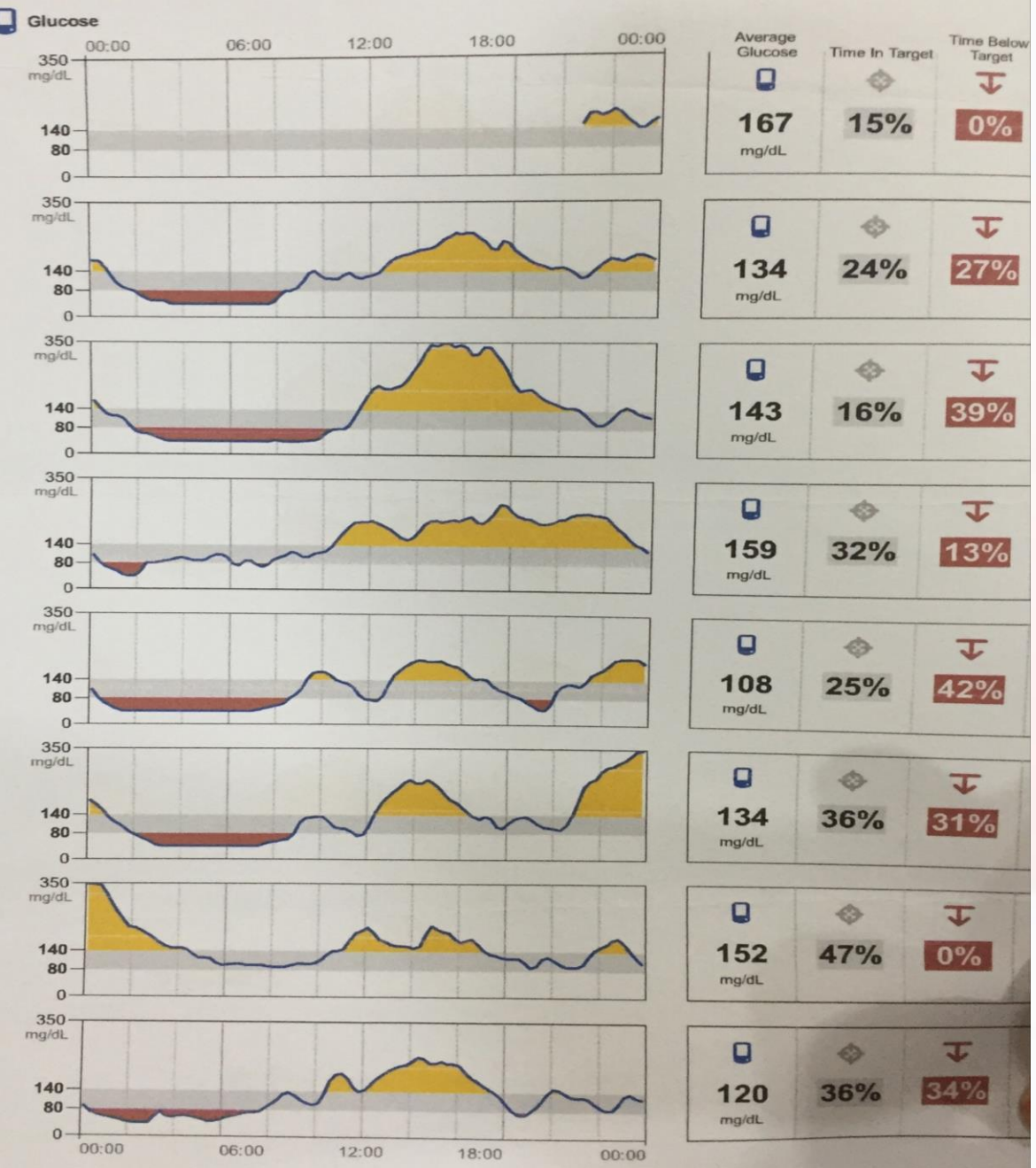
- Patients initiating insulin therapy should be aware of the potential for weight gain
- Major emphasis should be placed on diet and lifestyle modification to prevent it.
- The resulting weight gain worsens insulin resistance and may prompt insulin dose escalation, leading to a vicious cycle.
- Mutlifactorial
  - ✓ ***Continued dietary indiscretion.***
  - ✓ ***Conscious or subconscious snacking***
  - ✓ ***Overtreatment of hypoglycemia***

Dawn Phenomenon	Somogyi Effect
Normal morning effect ↑ growth hormones & ↓ insulin levels	Rebound hyperglycemia due to hypoglycemia throughout the night
Treatment Increasing medication (i.e. increase basal insulin) or adjust timing to improve control	**Treatment** Decreasing medication (i.e. decrease evening insulin) or adjust timing to prevent hypoglycemia

Kanika's Nursing Academy

How to differentiate: check blood sugar in middle of night (i.e. 0300). High/normal indicates Dawn Phenomenon, low indicates Somogyi effect





Average Glucose	Time In Target	Time Below Target
167 mg/dL	15%	0%
134 mg/dL	24%	27%
143 mg/dL	16%	39%
159 mg/dL	32%	13%
108 mg/dL	25%	42%
134 mg/dL	36%	31%
152 mg/dL	47%	0%
120 mg/dL	36%	34%

INSULIN +

METFORMIN

INSULIN SECRETOGOGUES

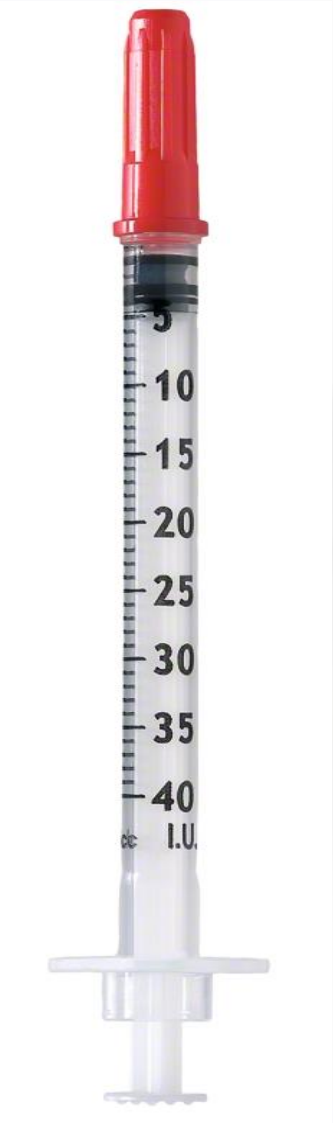
PIOGLITAZONE

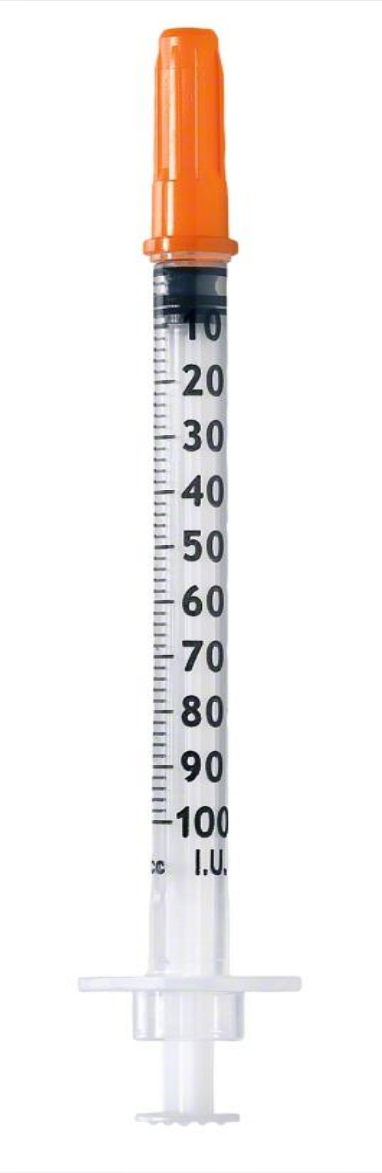
DPP-4 INHIBITORS

SGLT-2 INHIBITORS

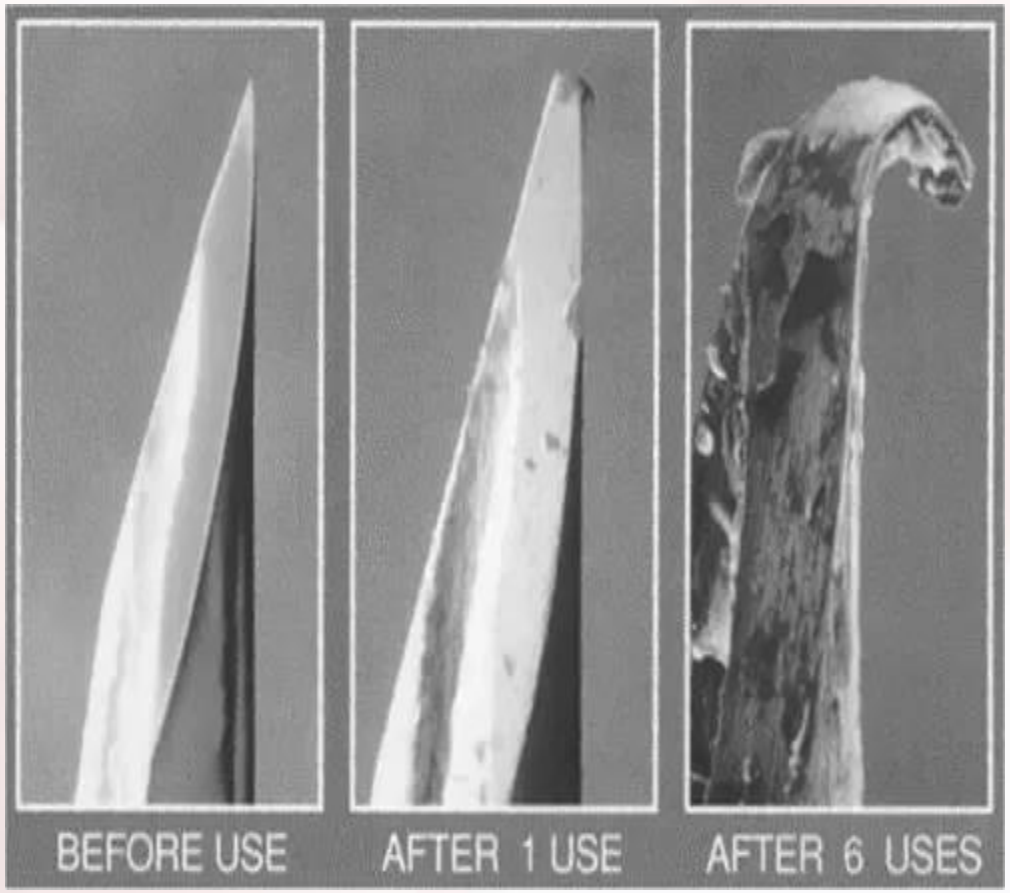
PRACTICAL ASPECTS OF INSULIN THERAPY

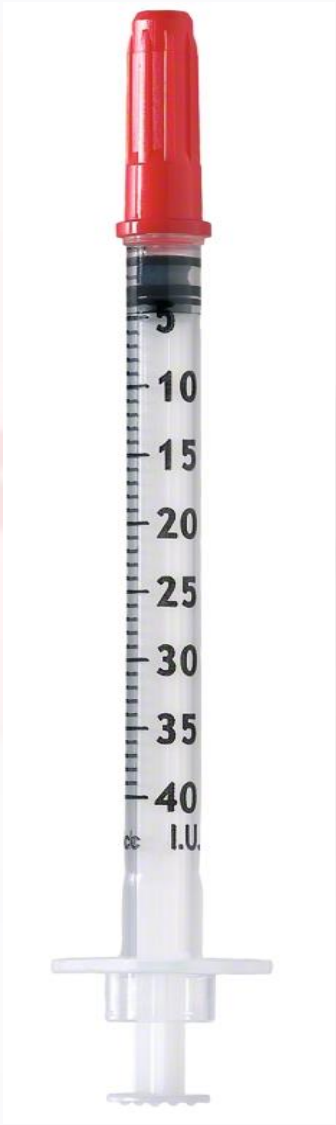
INSULIN SYRINGES

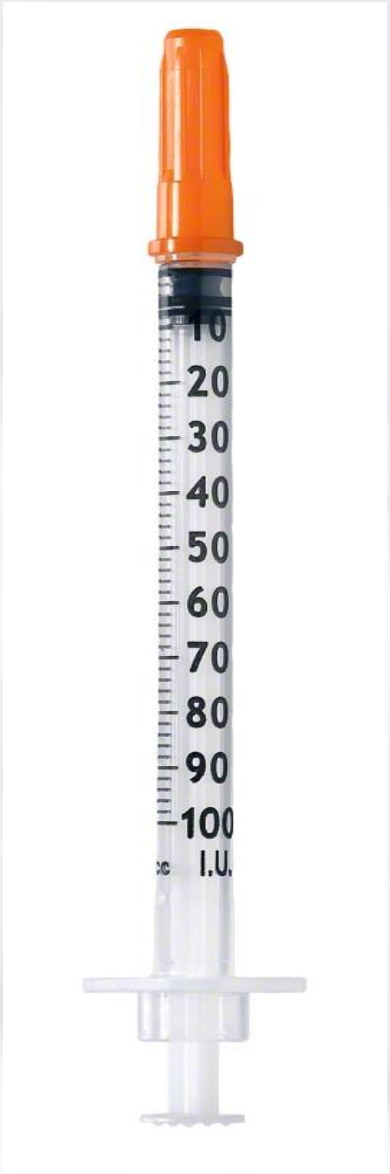




INSULIN SYRINGES







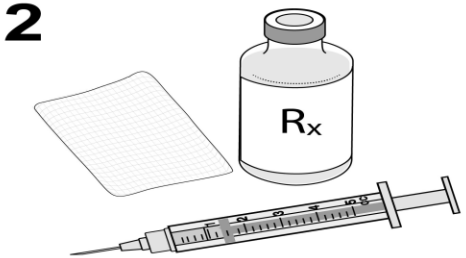
Steps to Drawing Up and Inject Insulin with a Syringe

1



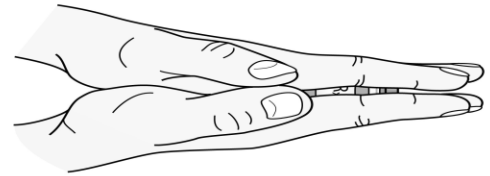
Always wash your hands.

2



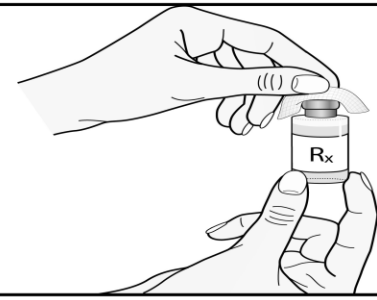
Get all of your supplies.

3



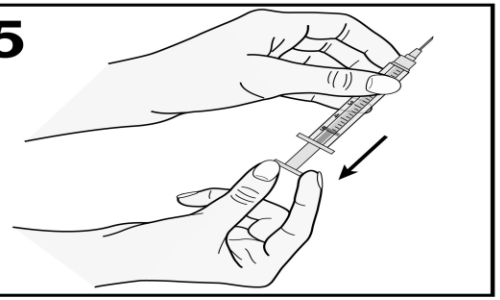
Roll the insulin vial between your hands, if using a cloudy insulin.

4



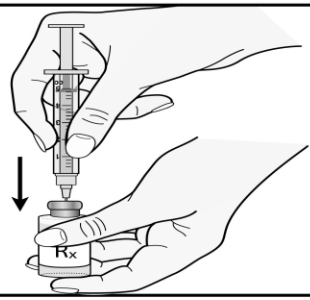
Wipe the top of the vial with an alcohol swab.

5



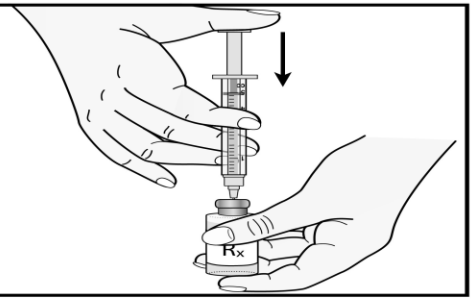
Pull the syringe plunger down to the desired units.

6



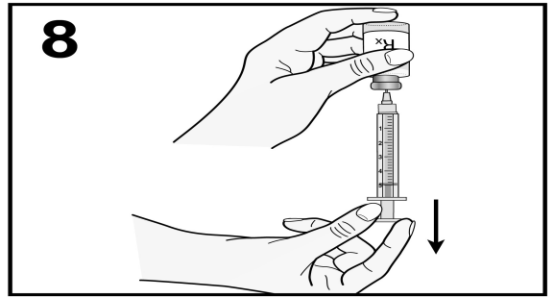
Push the syringe needle into the vial.

7



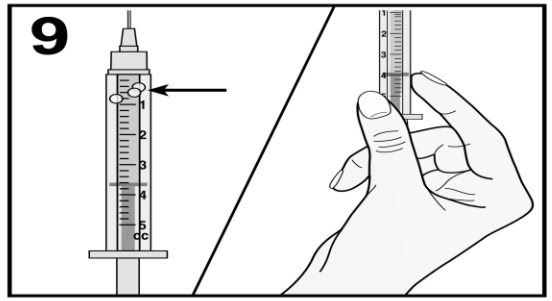
Push the plunger down.

8



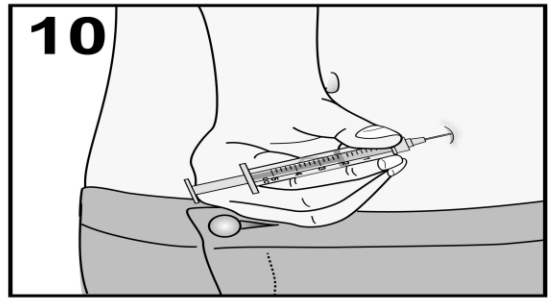
Turn the vial upside down. Pull the plunger down to withdraw insulin.

9



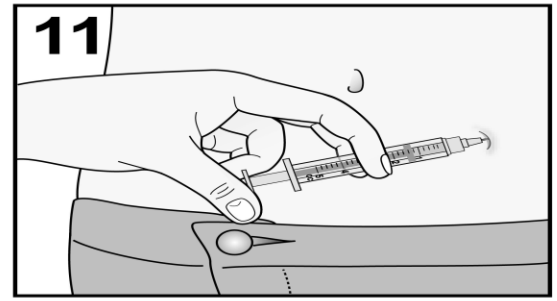
Remove all the air bubbles. Check to see if your dose is correct.

10

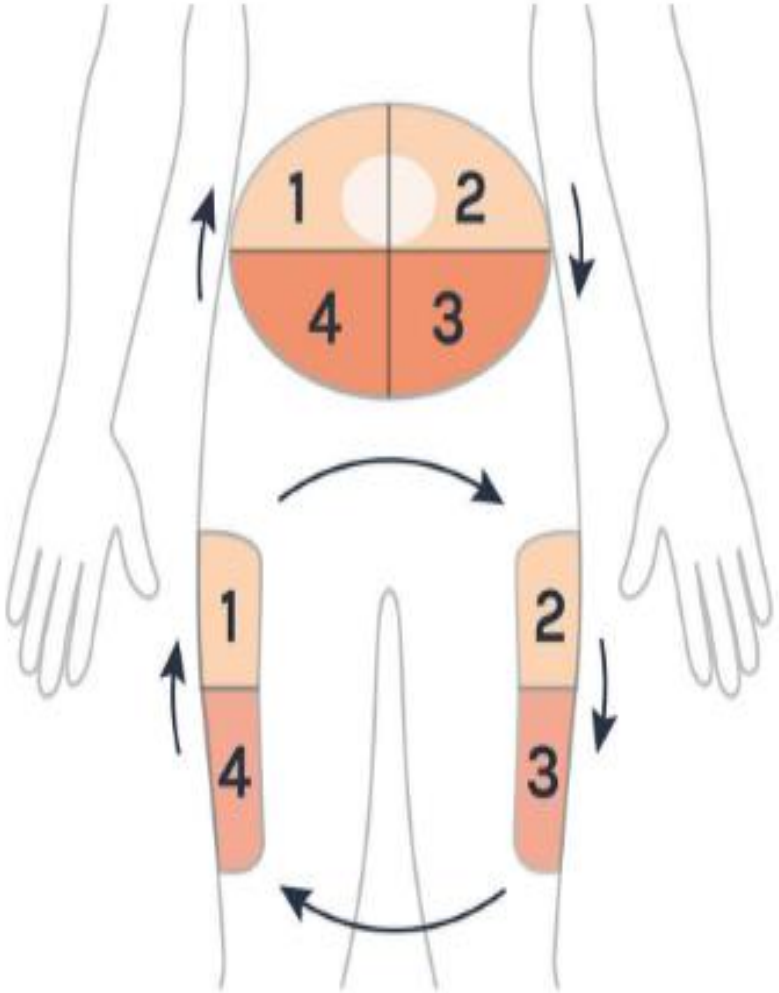


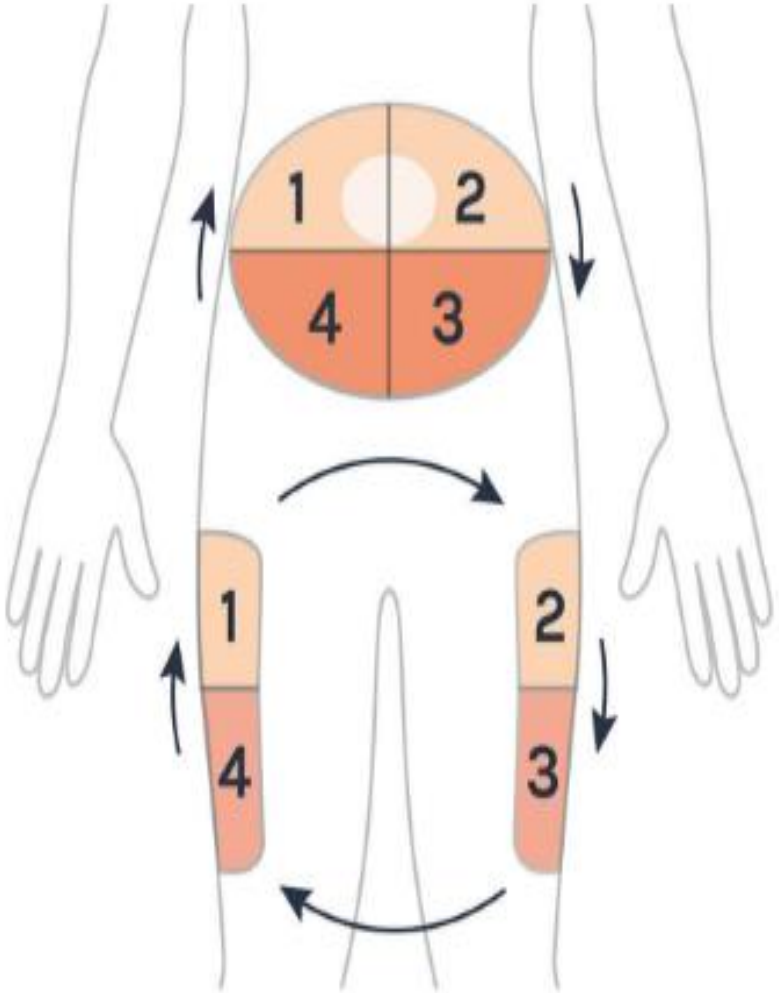
Pick a clean injection site. Push the needle into the skin.

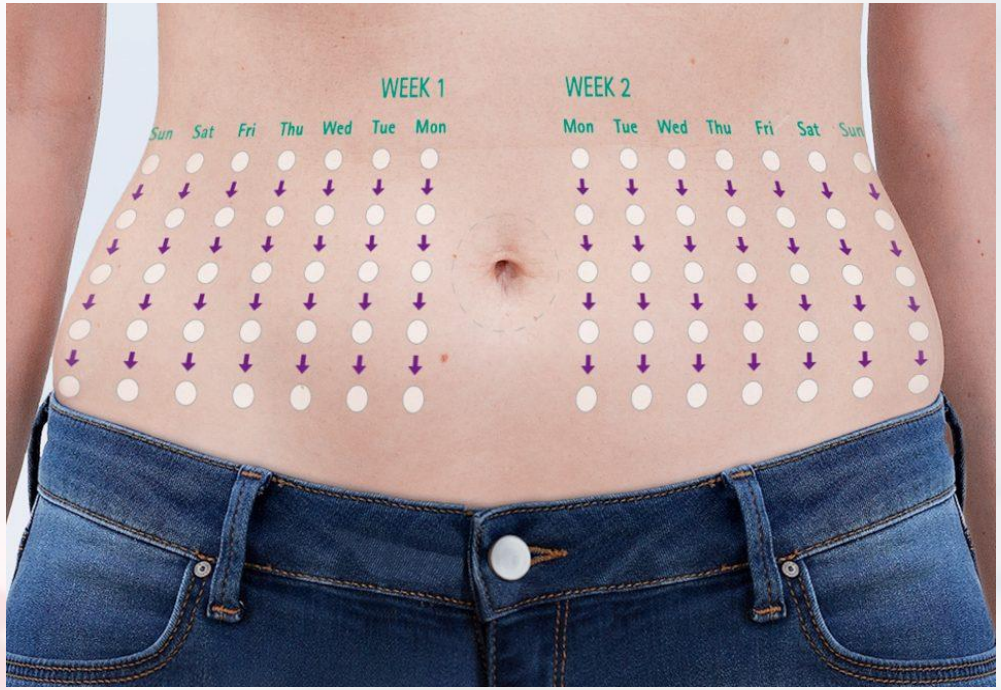
11

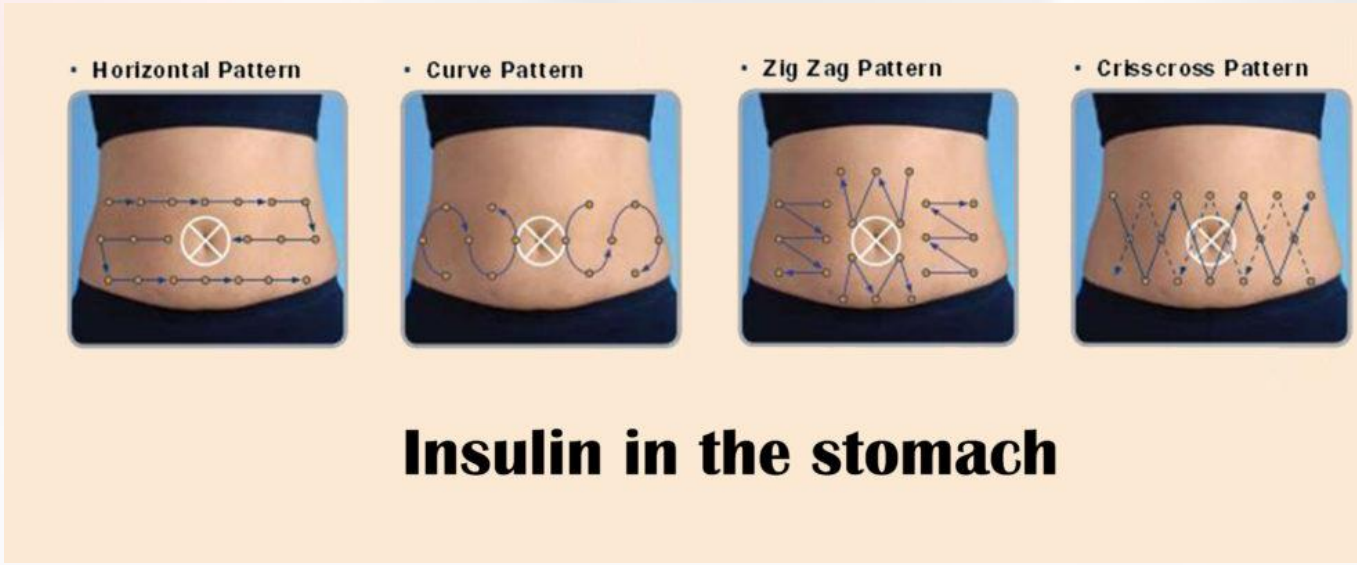


Push the plunger in. Pull the needle out of the skin.


1
2
4
3
1
2
4
3


1
2
4
3
1
2
4
3


WEEK 1
Sun Sat Fri Thu Wed Tue Mon
WEEK 2
Mon Tue Wed Thu Fri Sat Sun


• Horizontal Pattern
• Curve Pattern
• Zig Zag Pattern
• Crisscross Pattern
Insulin in the stomach

Insulin Storage

Unused insulin should be stored at 4-8°C in a refrigerator.

Insulin must never be frozen.

Direct sunlight or extreme heat can damage insulin.

After first usage, an insulin vial should be discarded after 3 months if kept at 2-8°C or 4 weeks if kept at room temperature.



International Diabetes Federation

INSULIN PEN DEVICES

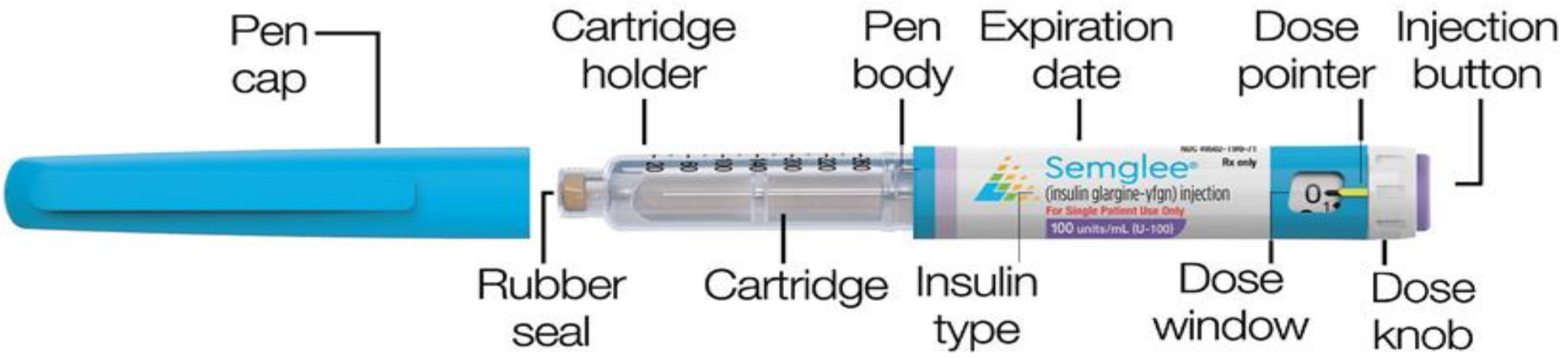
REUSABLE

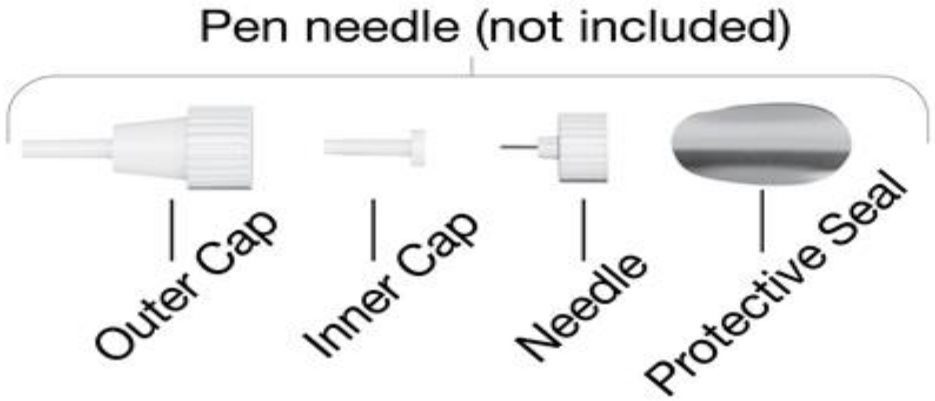


DISPOSABLE



INSULIN PEN DEVICES





CASE STUDIES

A 58 YEAR OLD GENTELMAN WORKING AS A LAWYER, KNOWN TO HAVE TYPE 2 DM FOR 10 YEARS, SYSTEMIC HYPERTENSION, DYSLIPIDEMIA.
IS ON ***TAB METFORMIN 1 g BD, TAB GLIMIPERIDE 4mg BD, AND SITAGLIPTIN 50mg OD, TAB TELMISARTAN 40MG OD, TAB ATORVASTATIN 10MG BEDTIME***.
HIS LABS SHOW ***FBS-210, PPBS-280, HbA1C IS 8.5%.***
WHAT WOULD YOU ADVICE...?

A 58 YEAR OLD GENTELMAN WORKING AS A LAWYER, KNOWN TO HAVE TYPE 2 DM FOR 10 YEARS, SYSTEMIC HYPERTENSION, DYSLIPIDEMIA.
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HIS LABS SHOW ***FBS-210, PPBS-280, HbA1C IS 8.5%.***
WHAT WOULD YOU ADVICE...?

- ✓ UNCONTROLLED DM DESPITE BEING ON MAXIMUM DOSE OF OADS.
- ✓ START AUGMENTATION THERAPY WITH BASAL INSULIN LIKE GLARGINE/DETEMIR
- ✓ START AT 10 UNITS BEDTIME OR 0.2 U/KG.
- ✓ IN ADDITION TO EXISTING OADS.
- ✓ EDUCATE ON SMBG AND TITRATE INSULIN DOSE TO TARGET FBS 80-120, AND PPBS 120-140.
- ✓ IF ON FOLLOW UP PPBS SHOW RISING TREND OR HBA1C REMAINS >7, PATIENT CAN TREATED WITH ORE-MIXED INSULIN AT 0.3 U/KG/DAY.

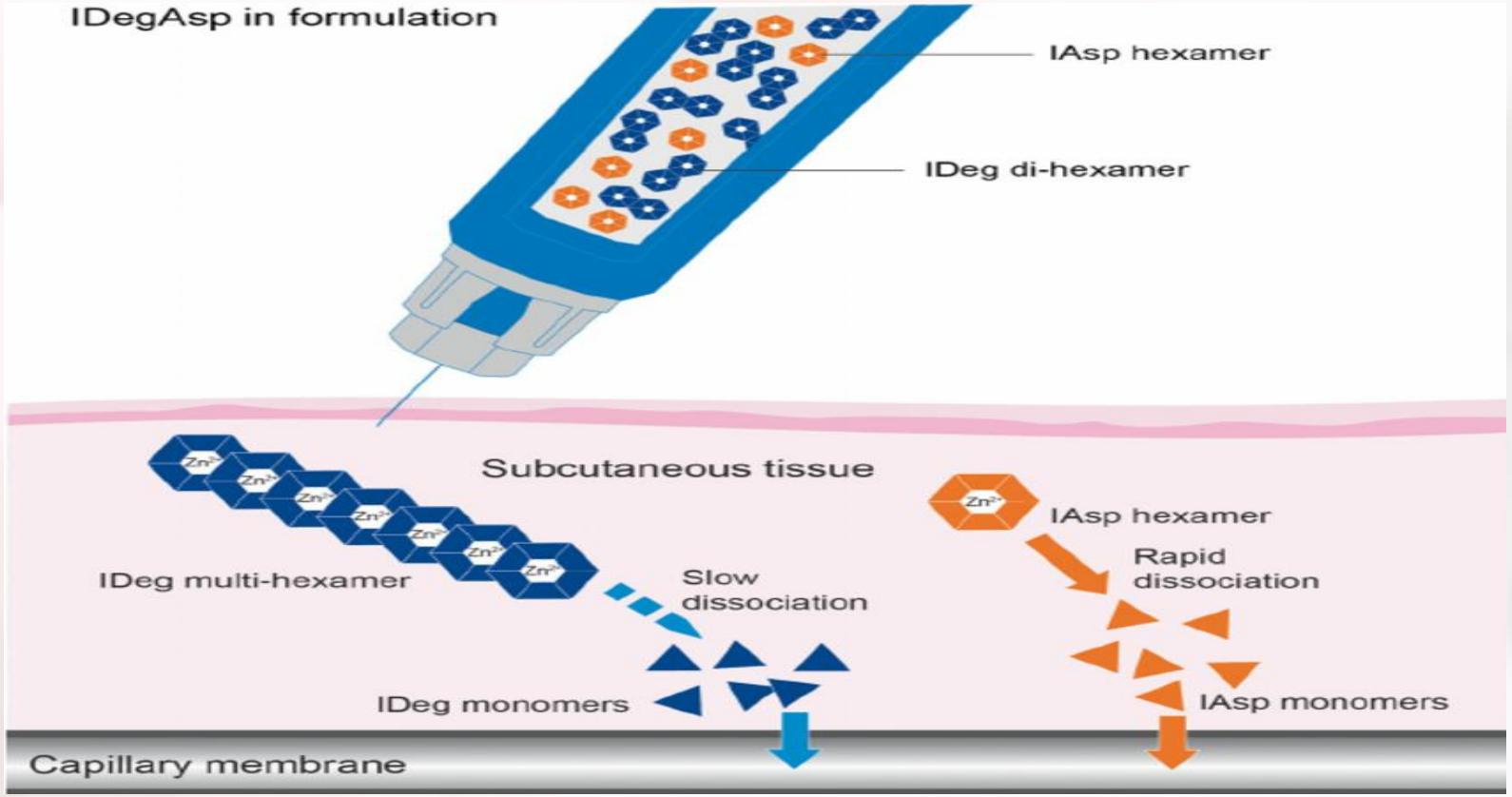
A 16 YEAR BOY WORKING AS A MASON PRESENTS WITH HISTORY OF POLYURIA, POLYDIPSIA, AND WEIGH LOSS OF 6KG OVER 3 MONTHS. HIS LABS SHOW FBG 280, PPBS 380, HbA1C 10.5%. GAD ANTIBODY IS POSITIVE. HOW WOULD YOU MANAGE..??

A 16 YEAR BOY WORKING AS A MASON PRESENTS WITH HISTORY OF POLYURIA, POLYDIPSIA, AND WEIGH LOSS OF 6KG OVER 3 MONTHS. HIS LABS SHOW FBG 280, PPBS 380, HbA1C 10.5%. GAD ANTIBODY IS POSITIVE. HOW WOULD YOU MANAGE..??

- ✓ PATIENT HAS TYPE 1 DM.
- ✓ REQUIRES REPLACEMENT INSULIN THERAPY AS BASAL-BOLUS REGIMEN.
- ✓ SHORT ACTING HUMAN INSULIN OR INSULIN ANALOGS AS BOLUS AND LONG ACTING INSULIN ANALOGS OR NPH FOR BASAL COVERAGE.
- ✓ HE CAN BE STARTED ON 0.5 U/KG/DAY, WHICH CAN BE SPLIT AS 40% BASAL AND 60% FPR PRANDIAL COVERAGE (20% BEFORE BREAKFAST, LUNCH & DINNER)
- ✓ EDUCATE ON DIETARY PLAN, PHYSICAL ACTIVITY, SMBG & HYPOGLYECEMIC SYMPTOMS.
- ✓ TITRATE INSULIN ON SMBG

NEWER ADVANCES IN INSULIN THERAPY

IDegAsp (70% degludec + 30% aspart)

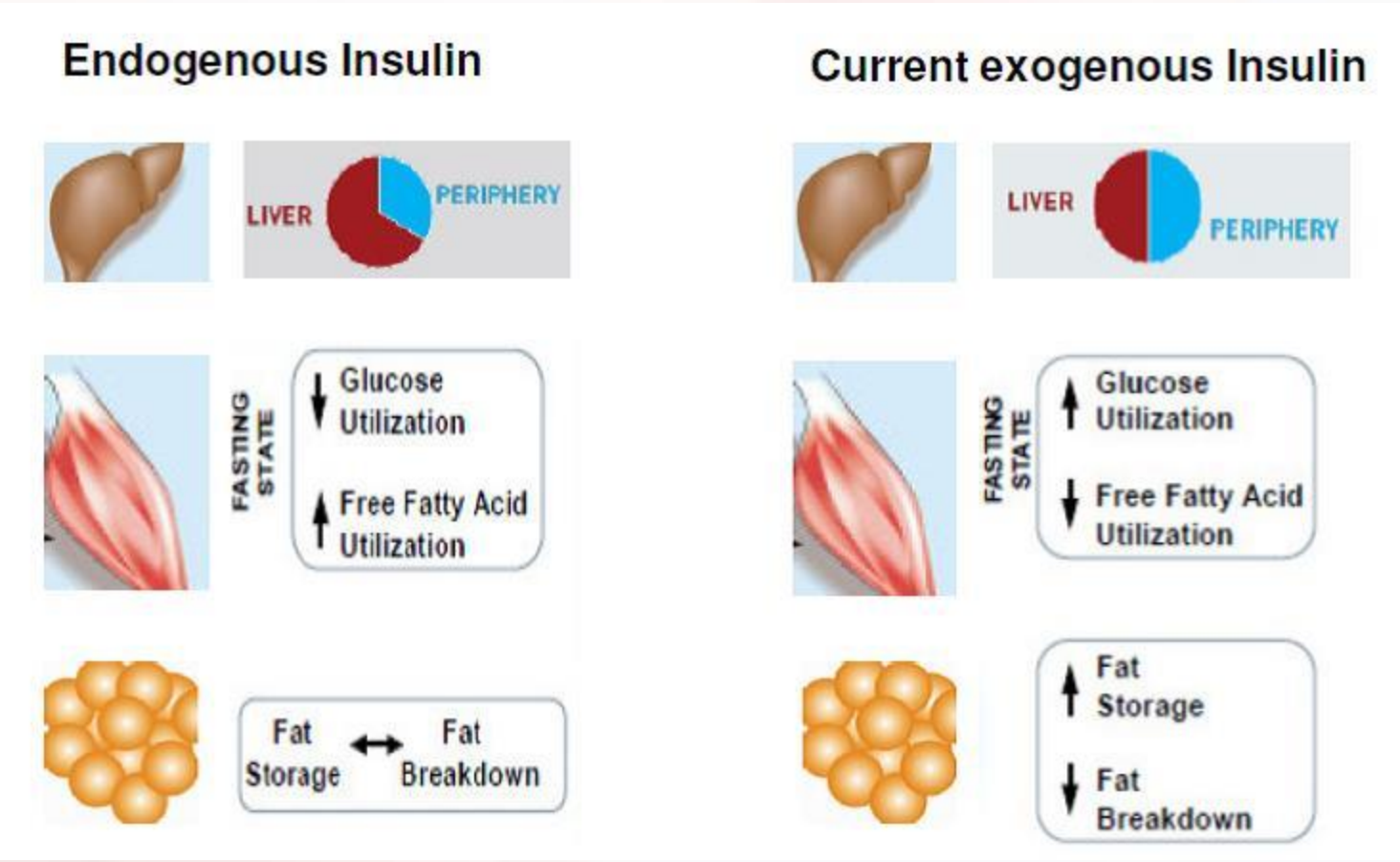


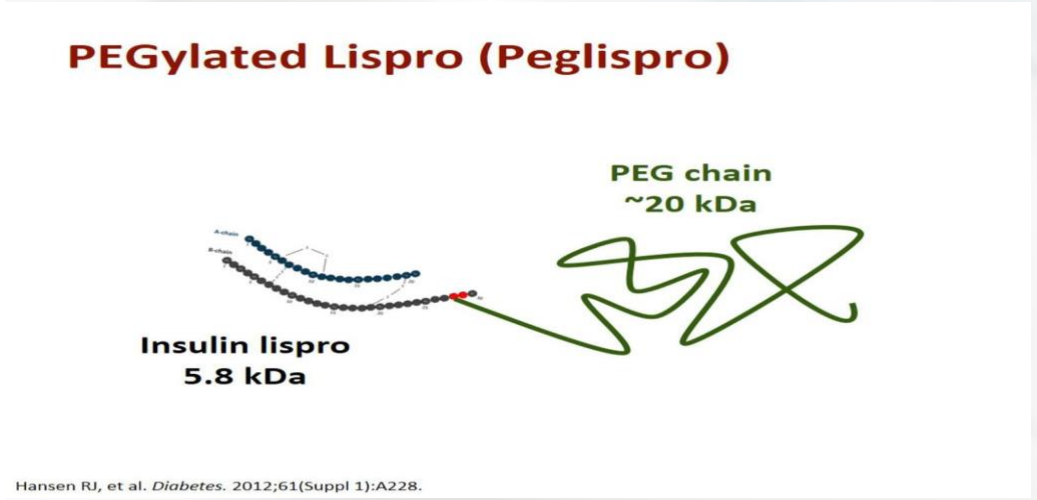


IDegLira (50U degludec + 1.8mg liraglutide)

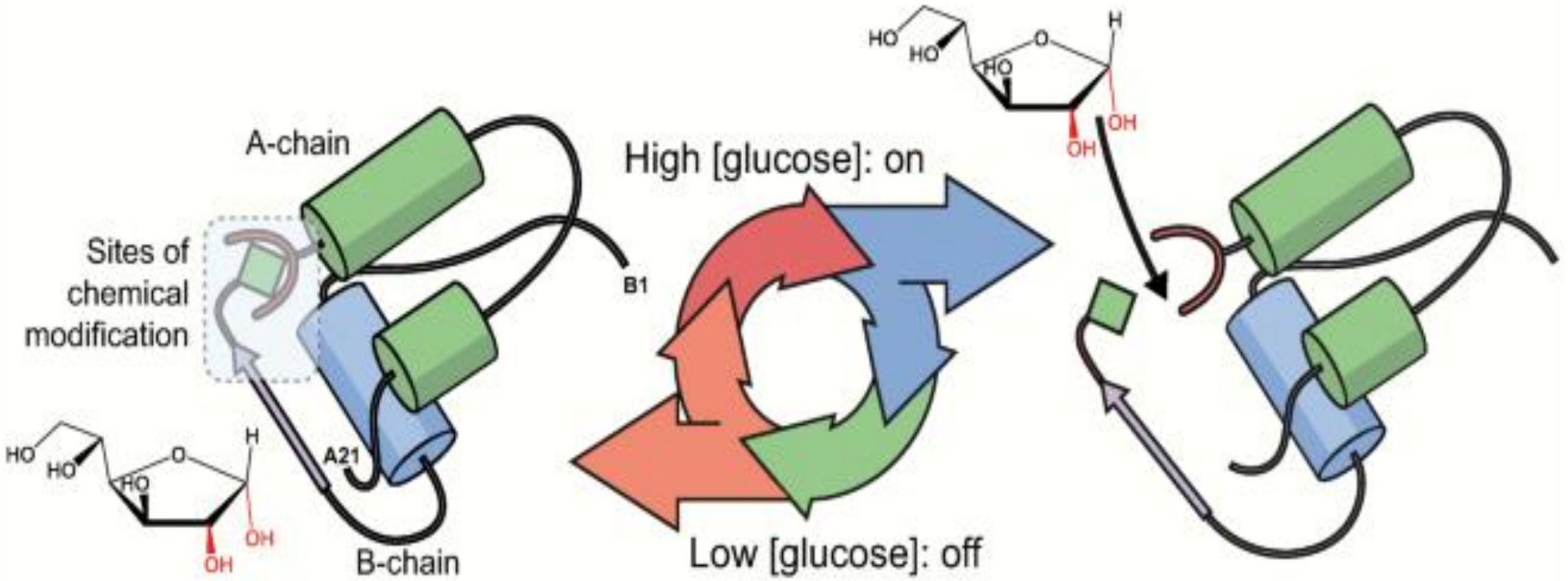


HEPATOSELECTIVE INSULIN





‘Smart’ insulin-delivery technologies and intrinsic glucose-responsive insulin analogues



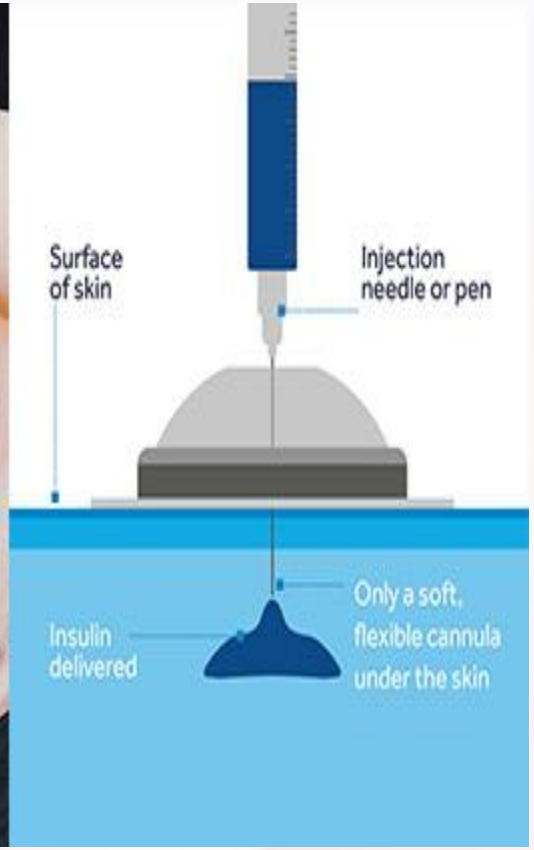
The Advantages of

INHALED INSULIN



Children with Diabetes®




Surface of skin
Injection needle or pen
Insulin delivered
Only a soft, flexible cannula under the skin



